

**O‘ZBEKISTON RESPUBLIKASI HARBIY XAVFSIZLIK VA MUDOFAA  
UNIVERSITETI  
AXBOROT-KOMMUNIKATSIYA TEXNOLOGIYALARI VA HARBIY  
ALOQA INSTITUTI**

**“KIBERXAVFSIZLIK” KAFEDRASI**

**O‘QUV QO‘LLANMA**

*“Kiber kriminalistika va etikal haking” fani*

**1-MAVZU  
WINDOWS OT XAVFSIZLIGI**

## **1-MAVZU**

### **WINDOWS OT XAVFSIZLIGI**

Ushbu qo‘llanmada “Windows OT xavfsizligi” mavzusining barcha to‘rtta mashg‘uloti (1-ma’ruza, 2-seminar, 3-amaliy, 4-amaliy) hech qanday oldingi bilimni talab qilmagan holda tushuntirilgan. Har bir bo‘lim: (1) nazariy tushuntirish, (2) tushunishni osonlashtiruvchi chizma/jadval, (3) amaliy ahamiyati va (4) nazorat savollari bilan yakunlanadi. Qo‘llanmani oxirigacha o‘qib chiqqan kursant Windows operatsion tizimining asosiy xavfsizlik mexanizmlarini (servislar, UAC, AppLocker, SmartScreen, Defender, Firewall, foydalanuvchi va xavfsizlik siyosatlarini) tushunadi va amalda sozlay oladi.

#### **Mavzuning umumiy tuzilishi**

- 1-mashg‘ulot (Ma’ruza, 2 soat): Servislar xususiyatlari, vazifalari va ularni ishga tushirish;
- 2-mashg‘ulot (Seminar, 2 soat): UAC, AppLocker, SmartScreen, Defender, Firewall xavfsizlik tizimlari;
- 3-mashg‘ulot (Amaliy, 4 soat): UAC, AppLocker, SmartScreen xavfsizlik tizimlarini amalda sozlash;
- 4-mashg‘ulot (Amaliy, 4 soat): Xavfsizlik sozlamalarini va foydalanuvchilar xavfsizligini sozlash.

**Nega bu mavzu muhim?** Kiberxavfsizlikda deyarli barcha hujum va himoya harakatlari operatsion tizim darajasida sodir bo‘ladi. Buzg‘unchi tizimga kirgach, u avvalo mavjud servis va xavfsizlik mexanizmlari bilan — UAC, Defender, Firewall bilan — to‘qnashadi; aksincha, himoyachi (Blue Team) va kiberkriminalist aynan shu mexanizmlarning loglari va sozlamalari orqali hujumni aniqlaydi. Shuning uchun bu mavzu keyingi barcha mavzular (virusologiya, tarmoq monitoringi, etikal haking) uchun poydevor hisoblanadi.

## 1-mashg'ulot. Windows OT servislari (Ma'ruza)

**Mavzu:** Windows OT xavfsizligi    **Mashg'ulot turi:** Ma'ruza    **Soat:** 2 soat

### O'quv savollari

1. Servislar xususiyatlari va vazifalari
2. Servislarni ishga tushirish

### 1-savol. Servislar xususiyatlari va vazifalari

#### 1.1. Windows servis (xizmat) nima?

Har qanday kompyuterda ikki xil dastur ishlaydi: birinchisi — siz sichqoncha bilan ochadigan, ekranda oynasi ko'rinadigan oddiy dasturlar (masalan, Word, brauzer). Ikkinchisi — sizga ko'rinmaydigan, ekranda oynasi bo'lmagan, lekin doimo fonda ishlab turadigan dasturlar. Ana shu ikkinchi turi servis (xizmat) deb ataladi.

Servisning oddiy dasturdan uchta asosiy farqi bor:

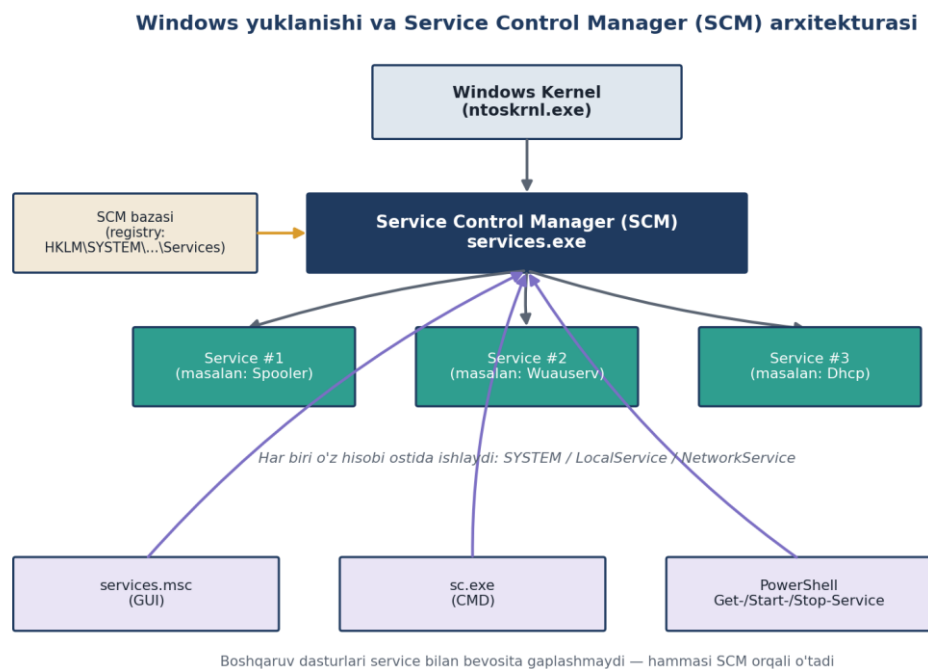
- **Foydalanuvchisiz ishlaydi:** servis tizimga hech kim login qilmagan bo'lsa ham ishlashi mumkin — masalan, kompyuterni yoqib, hech qanday hisobga kirmasangiz ham, fonda o'nlab servis ishlab turadi.
- **Avtomatik ishga tushadi:** Windows yuklanishi bilanoq, foydalanuvchi hech narsa qilmasdan turib ishga tushadi.
- **Maxsus tizim hisoblari ostida ishlaydi:** oddiy dastur sizning hisobingiz nomidan ishlasa, servis ko'pincha SYSTEM, LocalService yoki NetworkService kabi maxsus, yuqori imtiyozli hisoblar ostida ishlaydi.

#### 1.2. Nega bu kiberxavfsizlik uchun muhim?

Bu — butun mavzuning eng muhim nuqtasi va uni yaxshi tushunib olish kerak. Uch sababga ko'ra servislar xavfsizlik nuqtai nazaridan alohida e'tiborga loyiq:

- **Yuqori imtiyoz.** Ko'pgina servis SYSTEM darajasida ishlaydi — bu Windows'dagi eng yuqori imtiyoz darajasi bo'lib, hatto administratordan ham ko'proq huquqqa ega. Agar buzg'unchi shunday servisni nazoratga olsa, u orqali butun kompyuterni to'liq boshqarib olishi mumkin.
- **Persistence (mustahkamlanish) vositasi.** Zararli dastur (malware) kompyuterga bir marta kirgach, kompyuter qayta yoqilganda ham ishlashda davom etishni xohlaydi. Buning eng ishonchli yo'li — o'zini yangi servis sifatida ro'yxatdan o'tkazish, chunki servislar Windows yuklanishi bilan avtomatik ishga tushadi.

- **Privilege escalation (huquqni oshirish) yo‘li.** Agar biror servis noto‘g‘ri sozlangan bo‘lsa (masalan, uning fayliga oddiy foydalanuvchi ham yozishi mumkin bo‘lsa), past huquqli buzg‘unchi shu kamchilik orqali SYSTEM darajasigacha ko‘tarilishi mumkin. Bunday holatlar orasida eng mashhuri — “unquoted service path” zaifligi bo‘lib, u MITRE ATT&CK bazasida T1574.009 texnikasi sifatida rasman qayd etilgan.



1-rasm. Windows yuklanishi va Service Control Manager (SCM) arxitekturası

### 1.3. Service Control Manager (SCM) — barcha servislarning “nazoratchisi”

Windowsda yuzlab servis bor va ularning har birini alohida-alohida nazorat qilib bo‘lmaydi. Shuning uchun maxsus markaziy komponent mavjud — Service Control Manager (SCM), u services.exe fayli ko‘rinishida ishlaydi va Windows yuklanishi bilanoq avtomatik ishga tushadi. SCM quyidagi vazifalarni bajaradi:

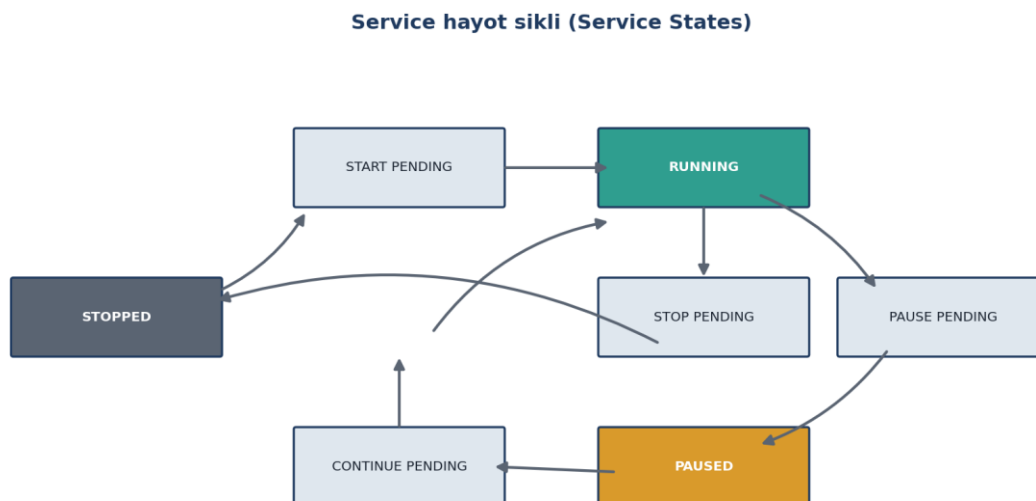
- O‘rnatilgan barcha servis va drayver- servislarning ma’lumotlar bazasini saqlaydi (bu ma’lumotlar Windows registrida, aniqrog‘i HKLM\SYSTEM\CurrentControlSet\Services kaliti ostida joylashgan).
- Har bir servisni tizim yuklanishida yoki so‘rov bo‘yicha ishga tushiradi.
- Ishlab turgan servislarning holatini (ishlayapti/to‘xtagan) doimiy kuzatib boradi.
- Har bir servis uchun kim unga kirish huquqiga ega ekanini (xavfsizlik tavsifi) belgilaydi.

Diqqat qilinsa, hech qaysi boshqaruv dasturi (services.msc, sc.exe, PowerShell) servis bilan bevosita gaplashmaydi — hammasi SCM orqali o‘tadi. Bu

markazlashgan arxitektura tufayli Windows servislarni masofadan turib ham xavfsiz boshqarish mumkin.

#### 1.4. Servisning hayot sikli (Service States)

Har bir servis o‘z hayoti davomida bir nechta holatdan (state) o‘tadi. Buni tushunish diagnostika va troubleshooting uchun muhim: masalan, agar biror servis uzoq vaqt “START PENDING” holatida qolib ketsa, bu muammo borligidan darak beradi.



2-rasm. Servis hayot sikli (Service States)

| Holat            | Ma’nosi                                                                                              |
|------------------|------------------------------------------------------------------------------------------------------|
| Stopped          | Servis to‘liq to‘xtagan, ishlamayapti.                                                               |
| Start Pending    | Servis ishga tushish jarayonida (hali to‘liq ishga tushmagan).                                       |
| Running          | Servis to‘liq ishlamoqda.                                                                            |
| Pause Pending    | Servis pauza qilinish jarayonida.                                                                    |
| Paused           | Servis vaqtincha to‘xtatilgan, lekin xotirada saqlanib turibdi va tezda qayta ishga tushishi mumkin. |
| Continue Pending | Pauzadan qayta ishga tushish jarayonida.                                                             |
| Stop Pending     | To‘xtatish jarayonida.                                                                               |

#### 1.5. Ishga tushirish turlari (Startup Type)

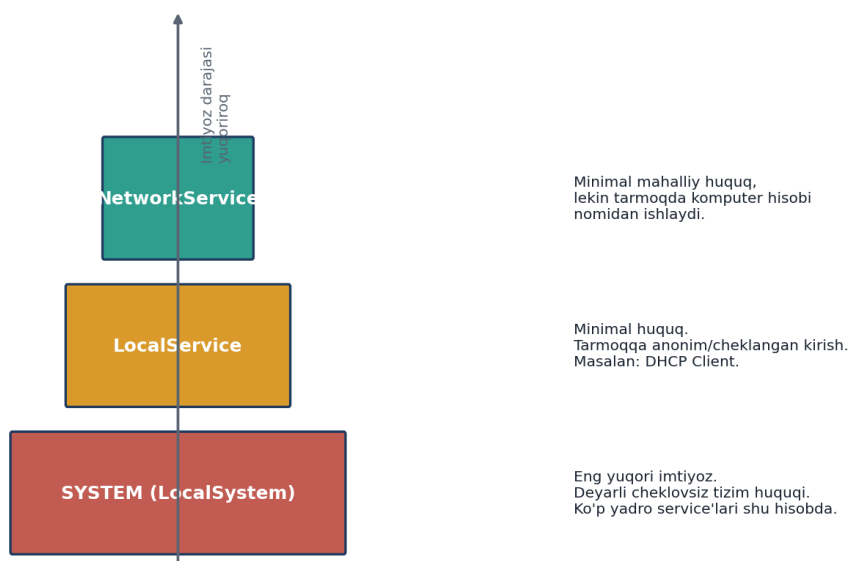
Har bir servis uchun u qachon ishga tushishi kerakligi alohida belgilanadi. Bu sozlama “startup type” deb ataladi va xavfsizlik nuqtai nazaridan juda muhim: ishlatilmayotgan servislarni “Disabled” holatiga o‘tkazish — hujum yuzasini (attack surface) kamaytirishning eng oddiy usullaridan biri.

| Turi                      | Ma'nosi                                                                   | Qachon ishlatiladi                                                                               |
|---------------------------|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Automatic                 | Tizim yuklanishi bilan darhol, boshqa hamma narsadan oldin ishga tushadi. | Doimiy kerak bo'lgan muhim servislar uchun (masalan, tarmoq, Windows Update).                    |
| Automatic (Delayed Start) | Tizim yuklanib bo'lgach, biroz kechikib ishga tushadi.                    | Zarur, lekin darhol shart bo'lmagan servislar uchun — bootni tezlashtiradi.                      |
| Manual                    | Faqat boshqa dastur yoki foydalanuvchi so'raganda ishga tushadi.          | Kamdan-kam ishlatiladigan servislar uchun (masalan, Print Spooler, agar printer kam ishlatilsa). |
| Disabled                  | Umuman ishga tushmaydi, hatto so'ralganda ham.                            | Ishlatilmaydigan, xavfsizlik nuqtai nazaridan xavfli bo'lishi mumkin bo'lgan servislar uchun.    |

### 1.6. Servis qaysi hisob ostida ishlaydi?

Har bir servis ma'lum bir foydalanuvchi hisobi “nomidan” ishlaydi va shu hisobning huquqlariga ega bo'ladi. Windows'da servislar uchun uchta maxsus, o'rnatilgan hisob mavjud:

Service hisoblari va imtiyoz darajasi



3-rasm. Servis hisoblari va ularning imtiyoz (privilege) darajasi

| Hisob                | Imtiyoz darajasi                | Tarmoqqa kirish          | Misol                              |
|----------------------|---------------------------------|--------------------------|------------------------------------|
| SYSTEM (LocalSystem) | Eng yuqori — deyarli cheklovsiz | Kompyuter hisobi nomidan | Ko'pgina yadro va tizim servisleri |
| LocalService         | Minimal, faqat mahalliy         | Anonim / juda cheklangan | DHCP Client, kabi past             |

|                |                                       |                          |                                        |
|----------------|---------------------------------------|--------------------------|----------------------------------------|
|                |                                       |                          | huquqli servislar                      |
| NetworkService | Minimal mahalliy, lekin tarmoqda faol | Kompyuter hisobi nomidan | Ba'zi tarmoq bilan ishlovchi servislar |

### Xavfsizlik nuqtai nazaridan yodda tuting

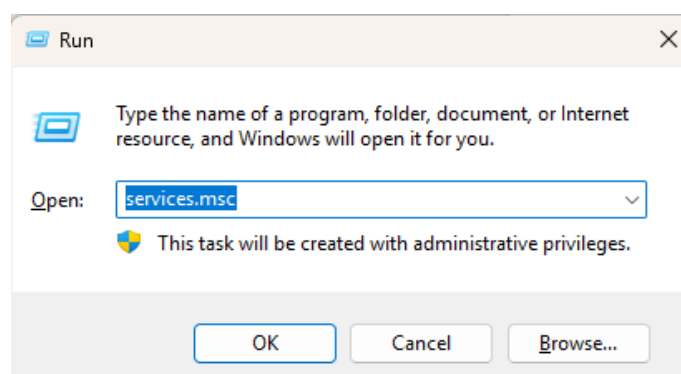
- SYSTEM huquqidagi servis — buzg'unchi uchun eng "qimmatli o'lja", chunki uni qo'lga kiritish butun tizimni boshqarish demakdir.
- Har doim "least privilege" (minimal zarur huquq) tamoyiliga amal qilish kerak: servisga qanchalik kam huquq berilgan bo'lsa, xavfsizlik shunchalik yuqori bo'ladi.

## 2-savol. Servislarni ishga tushirish

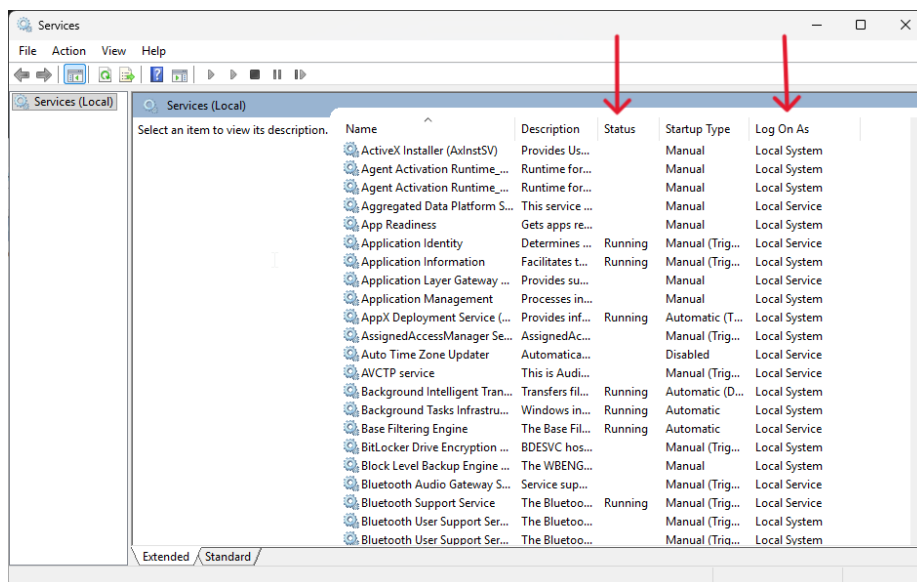
Windows servislarni boshqarish uchun uch xil asosiy vosita mavjud: grafik interfeys (services.msc), buyruq qatori vositasi (sc.exe) va PowerShell. Har uch vosita ham, yuqorida ko'rganimizdek, orqa fonda Service Control Manager bilan gaplashadi.

### 2.1. GUI orqali — services.msc

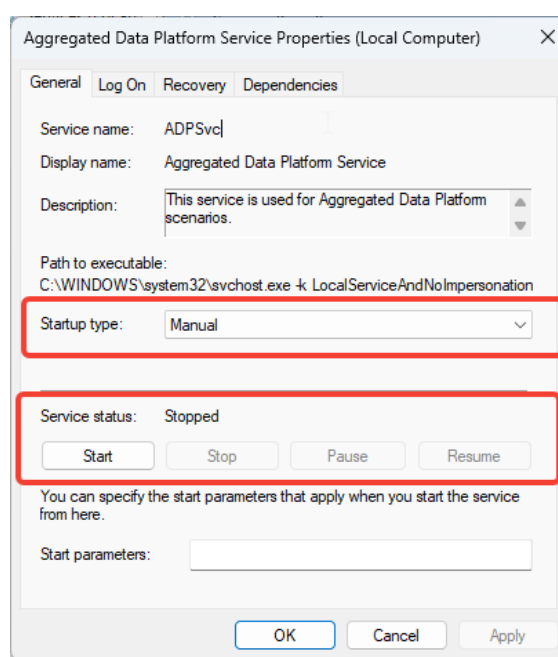
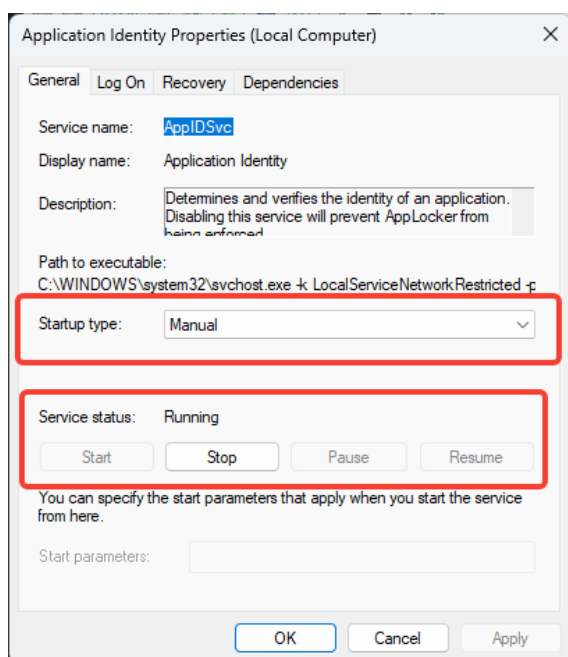
- Ochish uchun: klaviaturada **Win** + **R** tugmalarini bosing, ochilgan oynaga services.msc deb yozing va Enter bosing.



- Ochilgan oynada barcha servislar ro'yxati, ularning holati (Running / Stopped) va startup turi, "Log On" bo'limida qaysi hisob ostida ishlashi ko'rinadi.



- Biror servis ustiga ikki marta bosib (yoki o'ng tugma → Properties), uning: Startup type'ini o'zgartirish, servisni to'xtatish yoki ishga tushirish, ko'rish mumkin.



## 2.2. Buyruq qatori orqali — sc.exe

sc (Service Control) — buyruq qatoridan servislarni boshqarish uchun mo'ljallangan klassik Windows vositasi. Eng ko'p ishlatiladigan buyruqlar:

| Buyruq                 | Vazifasi                             | Oddiy user | Administrato<br>r |
|------------------------|--------------------------------------|------------|-------------------|
| sc query <servicename> | Servisning joriy holatini ko'rsatadi | Ishlaydi   | Ishlaydi          |
| sc start <servicename> | Servisni ishga tushiradi             | Ishlamaydi | Ishlaydi          |

| Buyruq                                        | Vazifasi                                                                                          | Oddiy user | Administrato<br>r |
|-----------------------------------------------|---------------------------------------------------------------------------------------------------|------------|-------------------|
| sc stop <servicename>                         | Servisni to'xtatadi                                                                               | Ishlamaydi | Ishlaydi          |
| sc config<br><servicename> start=<br>disabled | Startup turini "Disabled"ga<br>o'zgartiradi                                                       | Ishlamaydi | Ishlaydi          |
| sc qc <servicename>                           | Servis konfiguratsiyasini<br>(jumladan fayl yo'lini) ko'rsatadi<br>— xavfsizlik tekshiruvda muhim | Ishlaydi   | Ishlaydi          |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>C:\Users\user&gt;sc query Spooler</pre> <pre>SERVICE_NAME: Spooler         TYPE               : 110  WIN32_OWN_PROCESS (interactive)         STATE                : 4    RUNNING                         (STOPPABLE, NOT_PAUSABLE, IGNORES_SHUTDOWN)         WIN32_EXIT_CODE       : 0    (0x0)         SERVICE_EXIT_CODE   : 0    (0x0)         CHECKPOINT           : 0x0         WAIT_HINT            : 0x0</pre>                                                                                                                                                                                                                                                               | <pre>C:\Users\user&gt;sc stop Spooler</pre> <pre>SERVICE_NAME: Spooler         TYPE               : 110  WIN32_OWN_PROCESS (interactive)         STATE                : 1    STOPPED         WIN32_EXIT_CODE       : 0    (0x0)         SERVICE_EXIT_CODE   : 0    (0x0)         CHECKPOINT           : 0x0         WAIT_HINT            : 0x0</pre>                                                                                                                                                                                                                                                                                                                                      |
| <pre>C:\Users\user&gt;sc start Spooler</pre> <pre>SERVICE_NAME: Spooler         TYPE               : 110  WIN32_OWN_PROCESS (interactive)         STATE                : 2    START_PENDING                         (NOT_STOPPABLE, NOT_PAUSABLE, IGNORES_SHUTDOWN)         WIN32_EXIT_CODE       : 0    (0x0)         SERVICE_EXIT_CODE   : 0    (0x0)         CHECKPOINT           : 0x0         WAIT_HINT            : 0x7d0         PID                 : 4068         FLAGS                 :</pre>                                                                                                                                                                                | <pre>C:\Users\user&gt;sc config Spooler start= disabled</pre> <pre>[SC] ChangeServiceConfig SUCCESS</pre> <pre>C:\Users\user&gt;sc qc Spooler</pre> <pre>[SC] QueryServiceConfig SUCCESS</pre> <pre>SERVICE_NAME: Spooler         TYPE               : 110  WIN32_OWN_PROCESS (interactive)         START_TYPE          : 4    DISABLED         ERROR_CONTROL       : 1    NORMAL         BINARY_PATH_NAME    : C:\WINDOWS\System32\spoolsv.exe         LOAD_ORDER_GROUP    : SpoolerGroup         TAG                 : 0         DISPLAY_NAME        : Print Spooler         DEPENDENCIES        : RPCSS                         : http         SERVICE_START_NAME  : LocalSystem</pre> |
| <pre>C:\Users\user&gt;sc config Spooler start= auto</pre> <pre>[SC] ChangeServiceConfig SUCCESS</pre> <pre>C:\Users\user&gt;sc qc Spooler</pre> <pre>[SC] QueryServiceConfig SUCCESS</pre> <pre>SERVICE_NAME: Spooler         TYPE               : 110  WIN32_OWN_PROCESS (interactive)         START_TYPE          : 2    AUTO_START         ERROR_CONTROL       : 1    NORMAL         BINARY_PATH_NAME    : C:\WINDOWS\System32\spoolsv.exe         LOAD_ORDER_GROUP    : SpoolerGroup         TAG                 : 0         DISPLAY_NAME        : Print Spooler         DEPENDENCIES        : RPCSS                         : http         SERVICE_START_NAME  : LocalSystem</pre> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

### 2.3. PowerShell orqali

PowerShell — zamonaviy va avtomatlashtirish uchun eng qulay vosita. Asosiy buyruqlar (cmdlet'lar):

| Cmdlet                        | Vazifasi                               |
|-------------------------------|----------------------------------------|
| Get-Service                   | Barcha servislar ro'yxatini ko'rsatadi |
| Get-Service -Name "Spooler"   | Aniq bitta servis holatini ko'rsatadi  |
| Start-Service -Name "Spooler" | Servisni ishga tushiradi               |

| Cmdlet                                            | Vazifasi                    |
|---------------------------------------------------|-----------------------------|
| Stop-Service -Name "Spooler"                      | Servisni to'xtatadi         |
| Set-Service -Name "Spooler" -StartupType Disabled | Startup turini o'zgartiradi |

## 2.4. Bu bilim amalda qanday ishlatiladi?

Kiberkriminalistika va etikal haking nuqtai nazaridan bu bilim ikki tomonlama qo'llaniladi:

- **Himoya (Blue Team):** ortiqcha, ishlatilmayotgan servislarni aniqlab, ularni "Disabled" holatiga o'tkazish — hujum yuzasini kamaytiradi. Masalan, agar kompyuterda printer bo'lmasa, Print Spooler servisini o'chirib qo'yish tavsiya etiladi.
- **Hujum va tahlil (Red Team / Forensics):** kiberbuzg'unchi tizimga kirgach, ko'pincha yangi servis yaratib, o'zini shu orqali mustahkamlaydi. Shu sababli forensik tahlilda sc query yoki registrning Services kalitini tekshirish — birinchi qadamlardan biri hisoblanadi.

## Qo'shimcha muhim ma'lumot

Quyida amaliyotda tez-tez uchraydigan bir nechta muhim tizim servislari va ularning vazifasi keltirilgan — ularni bilish keyingi mashg'ulotlarda (ayniqsa xavfsizlik sozlamalarida) kerak bo'ladi:

| Servis nomi                        | Vazifasi                                                         |
|------------------------------------|------------------------------------------------------------------|
| Windows Update (wuauserv)          | Tizimga xavfsizlik va boshqa yangilanishlarni yuklab, o'rnatadi  |
| Print Spooler (spooler)            | Chop etish so'rovlarini navbatga qo'yib boshqaradi               |
| DHCP Client (dhcp)                 | Kompyuterga tarmoqdan avtomatik IP-manzil olishni ta'minlaydi    |
| DNS Client (dnscache)              | Domen nomlarini IP-manzilga aylantirish so'rovlarini keshlaydi   |
| Remote Procedure Call (RpcSs)      | Ko'p tizim komponentlari ishlashi uchun zarur asosiy servis      |
| Windows Defender Antivirus Service | Real vaqt rejimida zararli dasturlardan himoyani ta'minlaydi     |
| Windows Firewall (MpsSvc)          | Tarmoq trafigini nazorat qilib, xavfsizlik qoidalarini qo'llaydi |

## Nazorat savollari — 1-mashg'ulot

1. Windows servis oddiy dasturdan nimasi bilan farq qiladi?

2. SYSTEM, LocalService va NetworkService hisoblari o'rtasidagi farqni tushuntiring.
3. Service Control Manager (SCM) qanday vazifalarni bajaradi va u qayerda joylashgan?
4. Startup type'ning to'rtta turini sanab, har birining qachon ishlatilishini tushuntiring.
5. Nega noto'g'ri sozlangan servis orqali privilege escalation sodir bo'lishi mumkin?
6. sc query va sc config buyruqlari o'rtasidagi farq nima?
7. Nega forensik tahlilda servislarni tekshirish muhim hisoblanadi?

## 2-mashg'ulot. Windows OT servislari (Seminar)

**Mavzu:** Windows OT xavfsizligi    **Mashg'ulot turi:** Seminar    **Soat:** 2 soat

### O'quv savollari

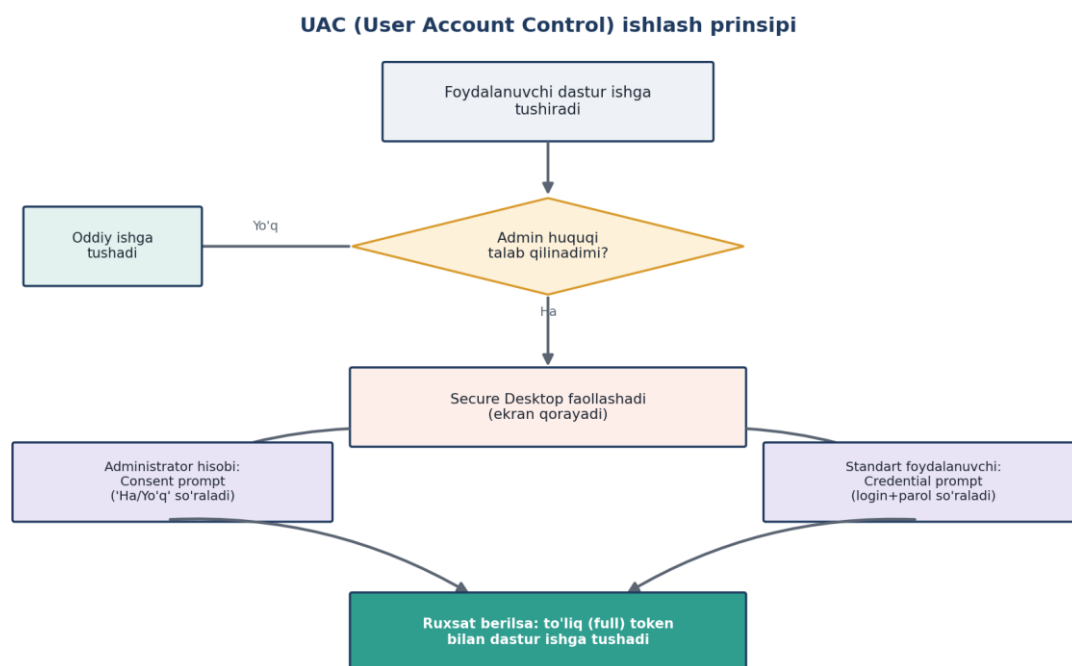
1. User account control, Applocker, smartscreen xavfsizlik tizimlari
2. Defender, firewall xavfsizlik tizimlari

### 1-savol. UAC, AppLocker, SmartScreen xavfsizlik tizimlari

#### 1.1. User Account Control (UAC)

Eski Windows versiyalarida (masalan, Windows XP) foydalanuvchilar ko'pincha doimiy ravishda administrator huquqi bilan ishlar edi. Bu juda xavfli edi: agar shunday foydalanuvchi tasodifan zararli dasturni ishga tushirib qo'ysa, u dastur ham darhol to'liq administrator huquqiga ega bo'lib qolar edi. UAC (User Account Control) - aynan shu muammoni hal qilish uchun Windows Vista'dan boshlab kiritilgan mexanizm.

UACning asosiy g'oyasi oddiy: hatto administrator hisobi ostida ishlayotgan bo'lsangiz ham, kundalik ishlaringizni tizim sizga "pasaytirilgan" (standart) huquq bilan bajartiradi. Agar biror dastur yoki amal administrator huquqini talab qilsa (masalan, dastur o'rnatish, tizim sozlamalarini o'zgartirish), ekranda maxsus so'rov - tasdiqlash oynasi chiqadi va faqat foydalanuvchi "Ha" desagina amal bajariladi.

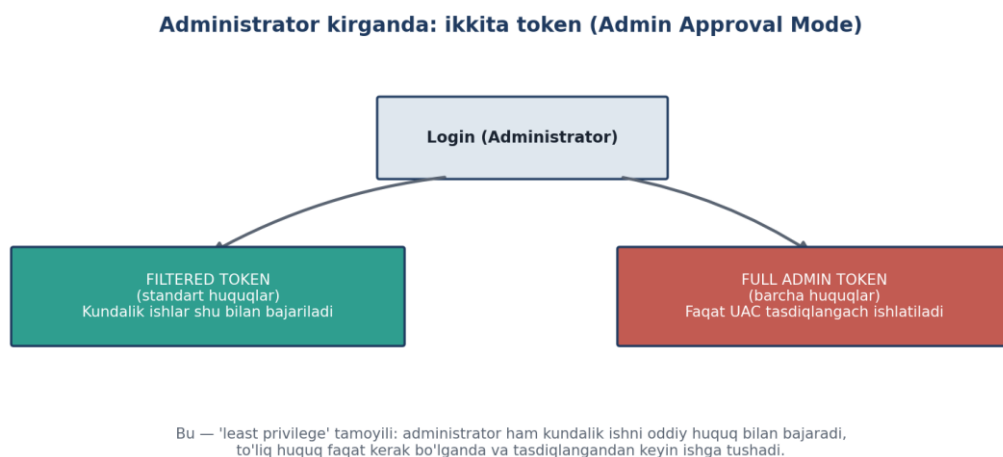


4-rasm. UAC (User Account Control) ishlash prinsipi

Diqqat qiling: tasdiqlash oynasi maxsus “Secure Desktop” rejimida ko‘rsatiladi (shuning uchun ekran birozgina qorayadi) - bu, zararli dastur soxta “Ha” tugmasini avtomatik bosib yubormasligi uchun qilingan xavfsizlik chorasi. So‘ralayotgan tasdiqning turi foydalanuvchi turiga bog‘liq:

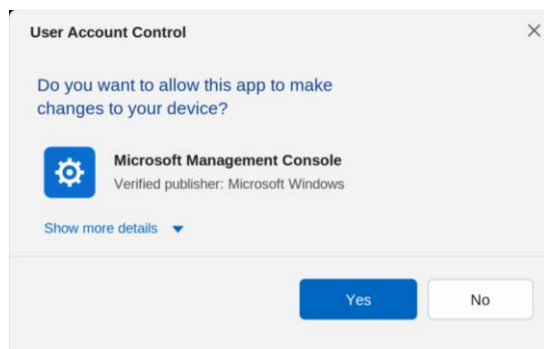
- **Administrator uchun — Consent prompt:** shunchaki “Ha” yoki “Yo‘q” tugmasini bosish kifoya.
- **Standart foydalanuvchi uchun — Credential prompt:** davom etish uchun administrator login va parolini kiritish talab qilinadi.

Bu qanday amalga oshiriladi? Administrator hisobi bilan tizimga kirganingizda, Windows sizga ikkita “token” (huquqlar to‘plami) beradi:



5-rasm. Administrator kirganda yaratiladigan ikkita token (Admin Approval Mode)

Bu — “least privilege” (minimal zarur huquq) tamoyilining amaliy ko‘rinishi: administrator ham kundalik ishni oddiy huquq bilan bajaradi, to‘liq huquq esa faqat kerak bo‘lgan va tasdiqlangan lahzada faollashadi.



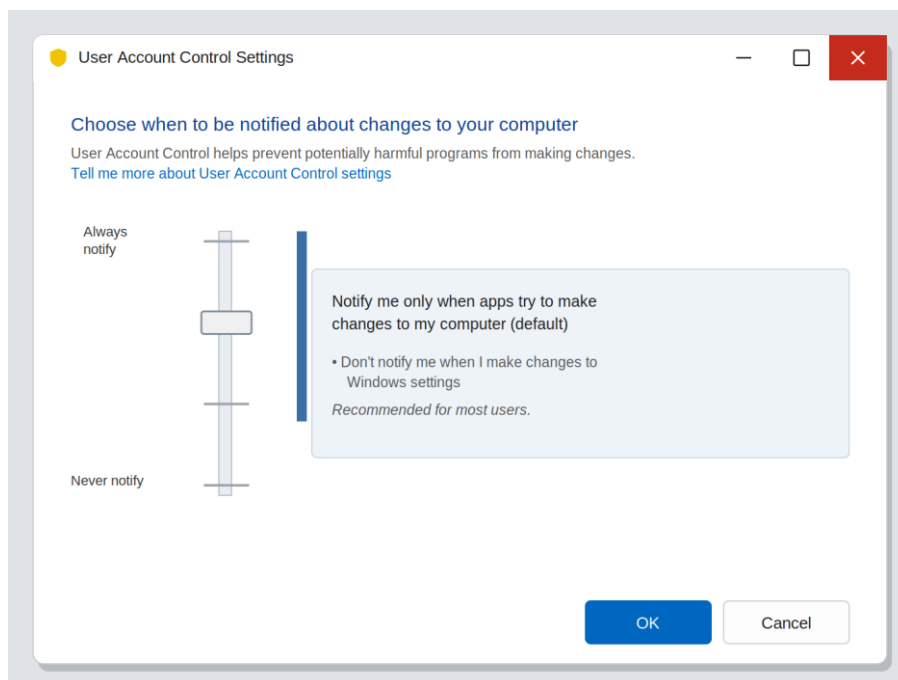
6-rasm. UAC tasdiqlash oynasi (Consent prompt) — Secure Desktop rejimida

Yuqoridagi oynada Windows dasturning nomini, uning nashriyotchisini (**Verified publisher**) va imzo holatini ko'rsatadi. Agar bu qatorda qizil rangda "Noma'lum nashriyotchi" (Unknown publisher) yozilgan bo'lsa — bu jiddiy ogohlantirish belgisi: imzolanmagan, ishonchsiz dastur administrator huquqini so'rayapti. Bunday paytda "Yo'q" tugmasini bosish xavfsizroq.

### *UAC darajalari (UAC slider)*

Boshqaruv panelida (Control Panel → User Accounts → Change User Account Control settings) UACning to'rtta sezuvchanlik darajasini sozlash mumkin:

| Daraja                                               | Tavsifi                                                                                                                                          |
|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Always notify                                        | Eng yuqori xavfsizlik: har qanday o'zgarishda (hatto Windows sozlamalarida ham) ogohlantiradi va Secure Desktop'ni ishlatadi.                    |
| Notify only when apps try to make changes (standart) | Faqat dasturlar o'zgarish kiritmoqchi bo'lganda ogohlantiradi; Windows sozlamalari uchun so'ramaydi. Ko'pchilik uchun tavsiya etiladigan daraja. |
| Notify only when apps try (ekranni qoraytirmasdan)   | Yuqoridagi kabi, lekin Secure Desktop ishlatilmaydi — xavfsizligi biroz pastroq.                                                                 |
| Never notify                                         | UAC to'liq o'chiriladi — tavsiya etilmaydi, xavfsizlikni sezilarli darajada pasaytiradi.                                                         |



7-rasm. UAC darajalarini sozlash oynasi (User Account Control Settings)

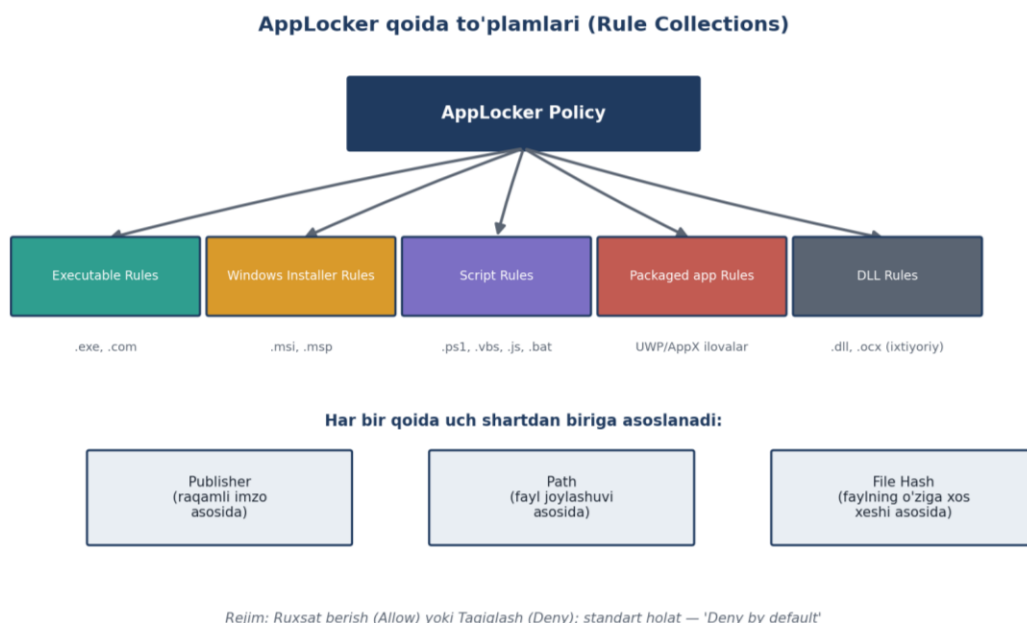
Amaliyotda ushbu slayder eng ko'p ishlatiladigan sozlamadir. Sukut bo'yicha u ikkinchi (standart) darajada turadi. Xavfsizlik talabi yuqori muhitlarda — masalan, harbiy yoki davlat tarmoqlarida — slayderni eng yuqori **Always notify** darajasiga

o‘tkazish tavsiya etiladi. Eng pastki **Never notify** darajasi UACni butunlay o‘chiradi va hech qachon ishlatilmasligi kerak.

## 1.2. AppLocker

UAC — kim nima qilishi mumkinligini nazorat qilsa, AppLocker — umuman qaysi dastur ishga tushishi mumkinligini nazorat qiladi. Boshqacha aytganda, AppLocker — “ruxsat etilgan dasturlar ro‘yxati” (whitelist) mexanizmi: administrator oldindan qaysi dasturlar, skriptlar yoki o‘rnatuvchilar ishga tushishi mumkinligini belgilaydi, qolgan hammasi avtomatik bloklanadi.

Bu, ayniqsa, korporativ va harbiy tashkilotlarda juda muhim: agar xodim tasodifan yoki ataylab noma’lum manbadan olingan zararli dastur (masalan, USB flesh orqali) ishga tushirmoqchi bo‘lsa, AppLocker uni avtomatik to‘xtatadi — hatto foydalanuvchi “Ha” desa ham.



### 8-rasm. AppLocker qoida to‘plamlari (Rule Collections) va shartlari

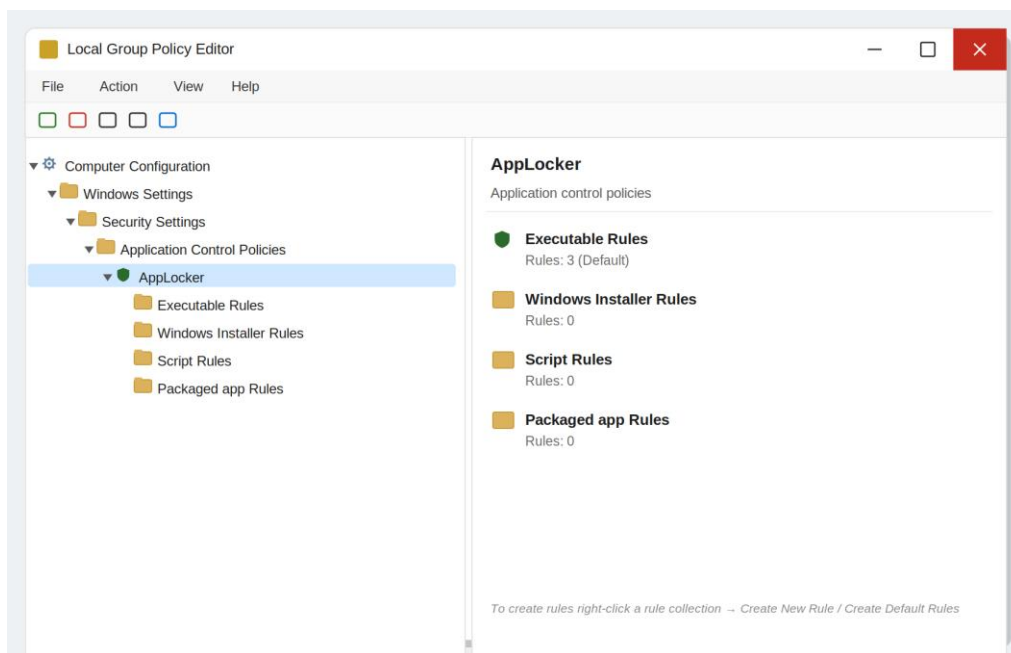
AppLocker beshta qoida to‘plamiga (Rule Collection) bo‘linadi — Executable (.exe/.com), Windows Installer (.msi/.msp), Script (.ps1/.vbs/.js/.bat), Packaged app (UWP ilovalar) va DLL. Har bir qoida esa uchta shartdan biriga asoslanishi mumkin:

- **Publisher:** dasturning raqamli imzosiga (ishlab chiqaruvchisiga) qarab ruxsat berish — eng moslashuvchan usul, chunki dastur yangilansa ham qoida ishlashda davom etadi.
- **Path:** faylning joylashgan manziliga qarab ruxsat berish (masalan, faqat C:\Program Files ichidagi dasturlarga ruxsat).

- **File Hash:** faylning aynan o‘ziga xos raqamli “barmoq izi”ga qarab ruxsat berish — eng qat’iy, lekin dastur yangilansa qoidani qayta yozish kerak bo‘ladi.

### Muhim eslatma

- AppLocker ishlashi uchun “Application Identity” (AppIDSvc) servisi ishga tushirilgan bo‘lishi shart — bu 1-mashg‘ulotda o‘rgangan servis tushunchasi bilan bevosita bog‘liq.
- AppLocker faqat Windows Enterprise/Education/Server nashrlarida to‘liq ishlaydi; Home va Pro nashrlarida cheklangan.

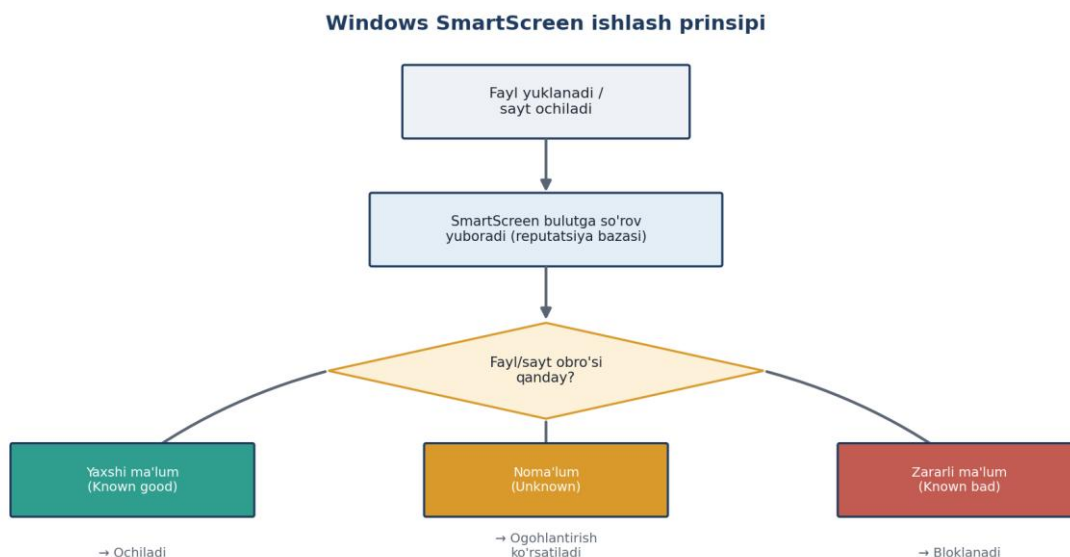


9-rasm. AppLocker qoidalari Local Group Policy Editor (gpedit.msc) ichida

Amaliy misol: agar tashkilotda faqat C:\Program Files va C:\Windows papkalaridan dastur ishga tushishiga ruxsat beruvchi Path qoidasi yaratilsa, foydalanuvchining Downloads papkasidan yoki USB fleshdan ishga tushirmoqchi bo‘lgan har qanday .exe fayli avtomatik bloklanadi — hatto foydalanuvchi administrator bo‘lsa ham.

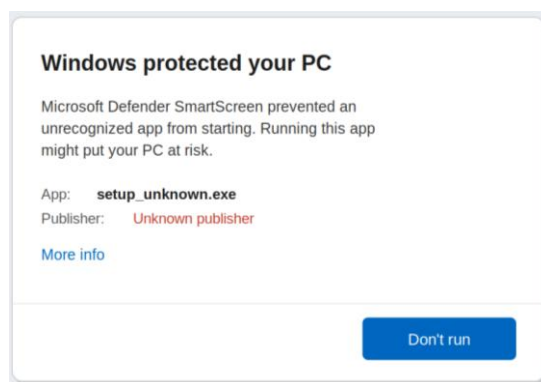
### 1.3. Windows SmartScreen

AppLocker mahalliy tarmoqdagi dasturlarni nazorat qilsa, SmartScreen — internetdan yuklab olingan fayllar va tashrif buyurilayotgan veb-saytlarning xavfliligini tekshiradi. SmartScreen Microsoft’ning bulutdagi katta ma’lumotlar bazasidan foydalanadi — bu bazada millionlab fayl va saytning “obro‘si” (reputatsiyasi) haqida ma’lumot to‘plangan.



*10-rasm. Windows SmartScreen ishlash prinsipi*

Fayl yoki sayt yangi bo‘lsa (hali hech kim ochmagan bo‘lsa) yoki obro‘si past bo‘lsa, SmartScreen “Noma‘lum nashriyotchi” yoki “yuklab olishni davom ettirasizmi?” kabi ogohlantirish ko‘rsatadi. Bu — phishing (firibgarlik) saytlari va yangi tarqatilayotgan zararli dasturlarga qarshi juda samarali himoya, chunki an‘anaviy antivirus imzosi hali mavjud bo‘lmagan tahdidlarni ham ushlab qolishi mumkin.



*11-rasm. Windows SmartScreen ogohlantirish oynasi*

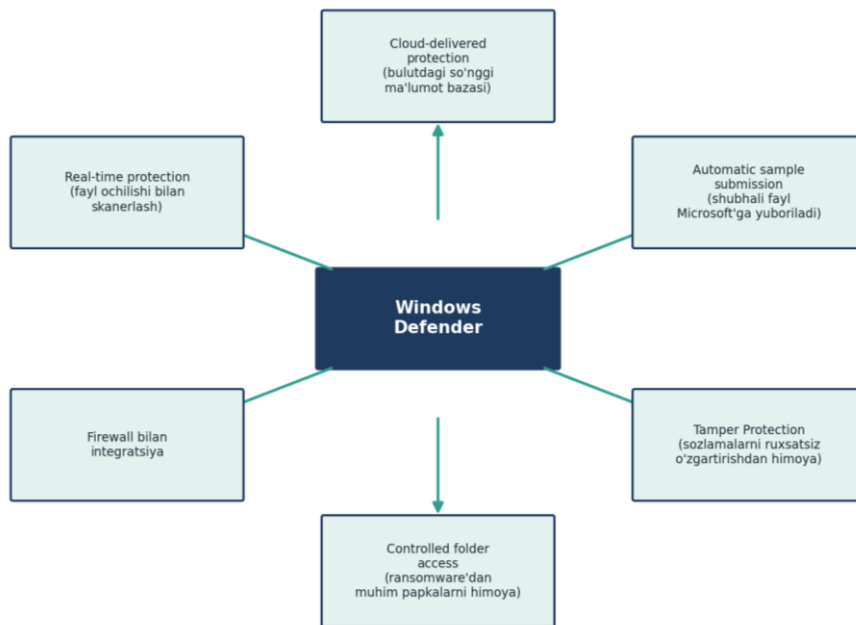
Masalan, internetdan yuklab olingan, imzolanmagan `setup_unknown.exe` faylini ishga tushirmoqchi bo‘lganingizda SmartScreen yuqoridagi oynani ko‘rsatadi va faylni bloklaydi. Foydalanuvchi faqat “More info → Run anyway” havolasi orqali javobgarlikni o‘z zimmasiga olib davom etishi mumkin — bu esa foydalanuvchini bir lahzaga to‘xtatib, o‘ylab ko‘rishga majbur qiladi.

## **2-savol. Defender, Firewall xavfsizlik tizimlari**

### **2.1. Windows Defender (Microsoft Defender Antivirus)**

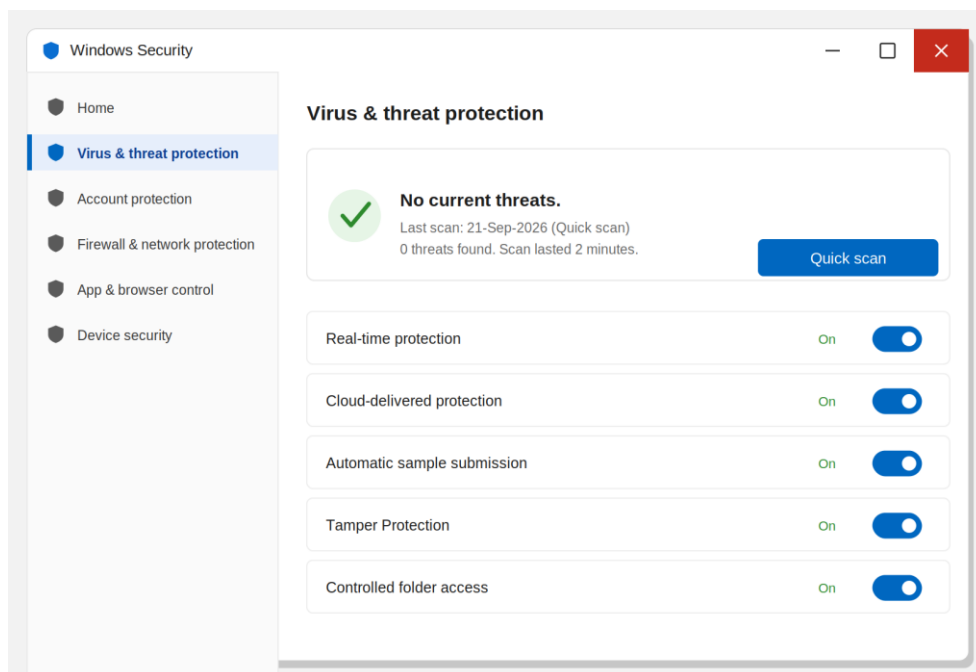
Windows Defender — Microsoft tomonidan yaratilgan, Windows’ga o‘rnatilgan holda keladigan antivirus va antimalware yechimi. UAC, AppLocker va SmartScreen “oldindan bloklashga” qaratilgan bo‘lsa, Defender — tizimda allaqachon mavjud yoki ishlayotgan fayllarni faol skanerlab, zararli xulq-atvorni aniqlashga qaratilgan.

**Windows Defender (Microsoft Defender Antivirus) komponentlari**



*12-rasm. Windows Defender komponentlari*

| Komponent                   | Vazifasi                                                                                                           |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------|
| Real-time protection        | Har bir fayl ochilishi, yuklanishi yoki ishga tushishi bilan oq uni darhol tekshiradi.                             |
| Cloud-delivered protection  | Mahalliy ma’lumotlar bazasida bo‘lmagan yangi tahdidlarni bulutdagi so‘nggi ma’lumot bazasi orqali aniqlaydi.      |
| Automatic sample submission | Shubhali, lekin hali noma’lum fayl namunasini tahlil uchun Microsoft’ga avtomatik yuboradi.                        |
| Tamper Protection           | Zararli dastur yoki buzg‘unchi Defender sozlamalarini o‘zgartirib, o‘chirib qo‘ymasligi uchun himoya qiladi.       |
| Controlled folder access    | Muhim papkalarni (Hujjatlar, Rasmlar) ruxsatsiz o‘zgartirishdan (masalan, ransomware shifrlashidan) himoya qiladi. |

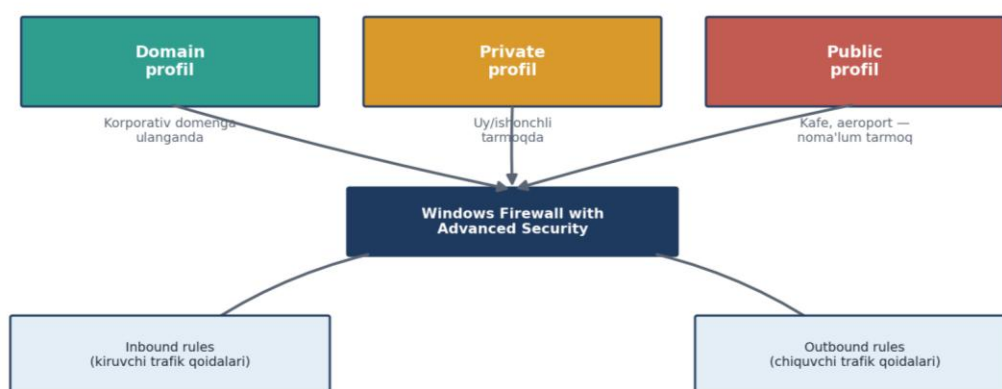


13-rasm. Windows Security — “Virus & threat protection” bo‘limi (Microsoft Defender)

## 2.2. Windows Firewall

Windows Firewall — kompyuter bilan tashqi tarmoq (internet yoki mahalliy tarmoq) o‘rtasidagi trafikni nazorat qiluvchi devor. U ikki yo‘nalishdagi trafikni tekshiradi: Inbound (kompyuterga kirayotgan) va Outbound (kompyuterdan chiqayotgan). Har bir qoida ma’lum port, dastur yoki protokol uchun trafikka ruxsat berish yoki uni bloklashni belgilaydi.

Windows Firewall — tarmoq profillari va qoida yo‘nalishi

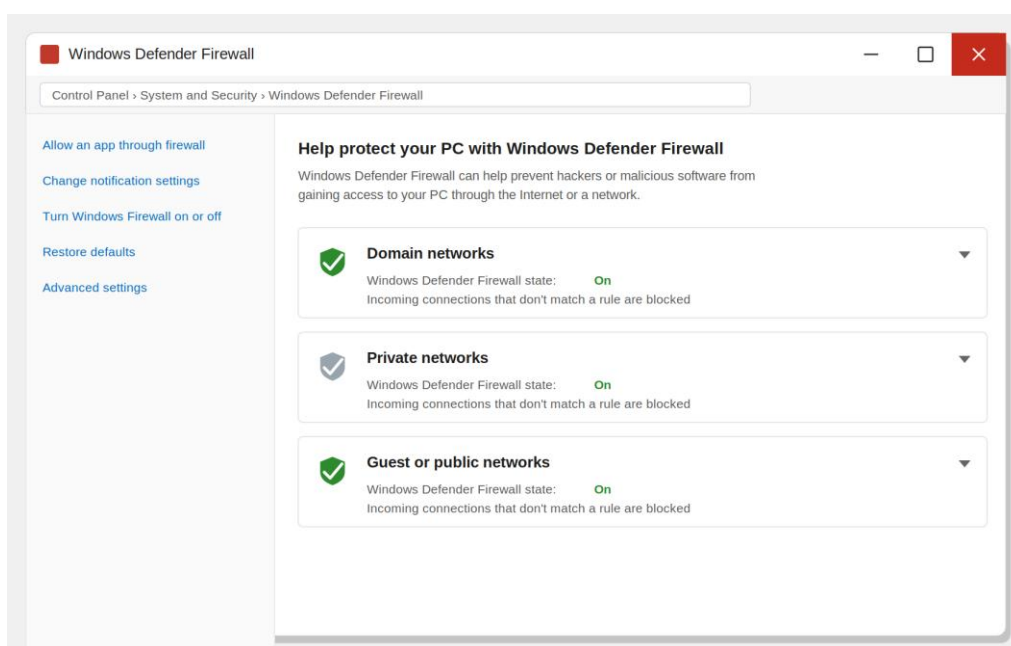


Har bir profil uchun alohida qoidalar to‘plami sozlanishi mumkin  
(masalan: Public profilda ko‘proq portlar yopiq bo‘ladi)

14-rasm. Windows Firewall tarmoq profillari va qoida yo‘nalishi

Firewall'ning muhim xususiyati — u qaysi tarmoq turiga ulanganingizga qarab turli profil qo‘llaydi, chunki uy tarmog‘ida xavf darajasi kafedagi ochiq Wi-Fi’dan farq qiladi:

| Profil  | Qachon qo‘llaniladi                                            | Odatiy xavfsizlik darajasi                           |
|---------|----------------------------------------------------------------|------------------------------------------------------|
| Domain  | Kompyuter korporativ domenga (Active Directory) ulangan bo‘lsa | O‘rtacha — ichki tarmoqqa ishonch yuqoriroq          |
| Private | Uy yoki ishonchli, shaxsiy tarmoqqa ulanganda                  | O‘rtacha                                             |
| Public  | Kafe, aeroport, mehmonxona kabi notanish tarmoqlarda           | Eng qattiq — ko‘pgina kiruvchi ulanishlar bloklanadi |



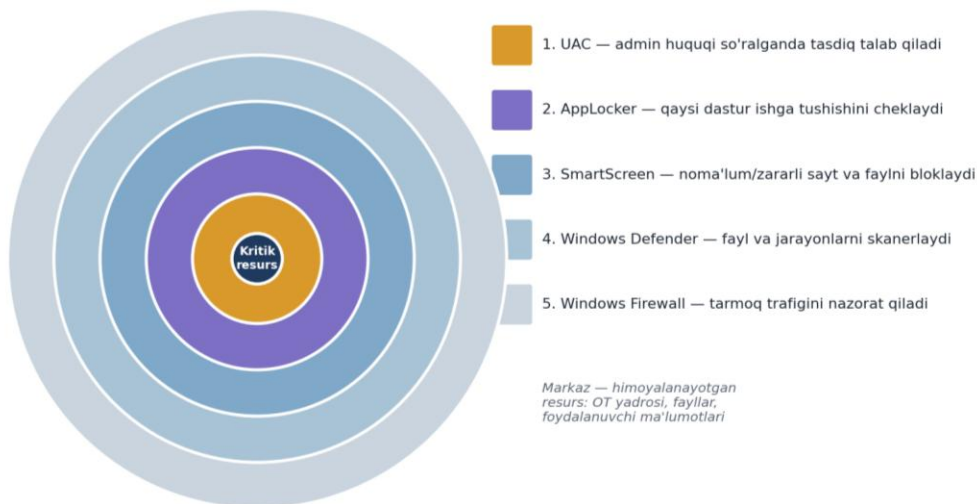
15-rasm. Windows Defender Firewall — tarmoq profillari (Control Panel)

Har bir profil uchun alohida qoidalar to‘plami saqlanadi. Masalan, kafedagi ochiq Wi-Fi’ga ulanganda Windows Public profilini qo‘llaydi va deyarli barcha kiruvchi ulanishlarni bloklaydi; uyingizdagi ishonchli tarmoqda esa Private profil bilan fayl almashish kabi xizmatlarga ruxsat beriladi.

### Qo‘shimcha muhim ma’lumot — barcha vositalarning birgalikdagi ishi

UAC, AppLocker, SmartScreen, Defender va Firewall — alohida-alohida ishlaydigan, lekin birgalikda “chuqur himoya” (defense in depth) tamoyilini tashkil etuvchi qatlamlardir. Bu tamoyilning mohiyati: agar buzg‘unchi bitta qatlamni chetlab o‘tishga muvaffaq bo‘lsa ham, keyingi qatlam uni ushlab qoladi.

### Windows OT: chuqur himoya qatlamlari (Defense in Depth)



16-rasm. Windows OT: chuqur himoya qatlamlari (Defense in Depth)

| Vosita           | Asosan nimadan himoya qiladi                                                              |
|------------------|-------------------------------------------------------------------------------------------|
| UAC              | Foydalanuvchi bilmagan holda admin huquqi ishlatilishidan                                 |
| AppLocker        | Ruxsat etilmagan/noma'lum dasturlarning ishga tushishidan                                 |
| SmartScreen      | Internetdan yuklangan zararli fayl va phishing saytlaridan                                |
| Windows Defender | Kompyuterda allaqachon mavjud yoki ishlayotgan zararli kod (virus, trojan, ransomware)dan |
| Windows Firewall | Tarmoq orqali keladigan yoki ketayotgan ruxsatsiz ulanishlardan                           |

### Nazorat savollari — 2-mashg'ulot

1. UAC nima uchun kiritilgan va u qanday muammoni hal qiladi?
2. Consent prompt va Credential prompt o'rtasidagi farq nima?
3. Filtered token va full admin token nima uchun kerak?
4. AppLocker'ning uchta qoida shartini (Publisher, Path, File Hash) tushuntiring.
5. SmartScreen qaysi turdagi tahdidlarga qarshi ayniqsa samarali?
6. Windows Defender'ning Controlled folder access komponenti nimadan himoya qiladi?
7. Domain, Private va Public firewall profillari orasidagi farqni tushuntiring.
8. "Defense in depth" (chuqur himoya) tamoyili nimani anglatadi?

### 3-mashg'ulot. Windows OT servislari (Amaliy)

**Mavzu:** Windows OT xavfsizligi    **Mashg'ulot turi:** Amaliy    **Soat:** 4 soat

#### O'quv savollari

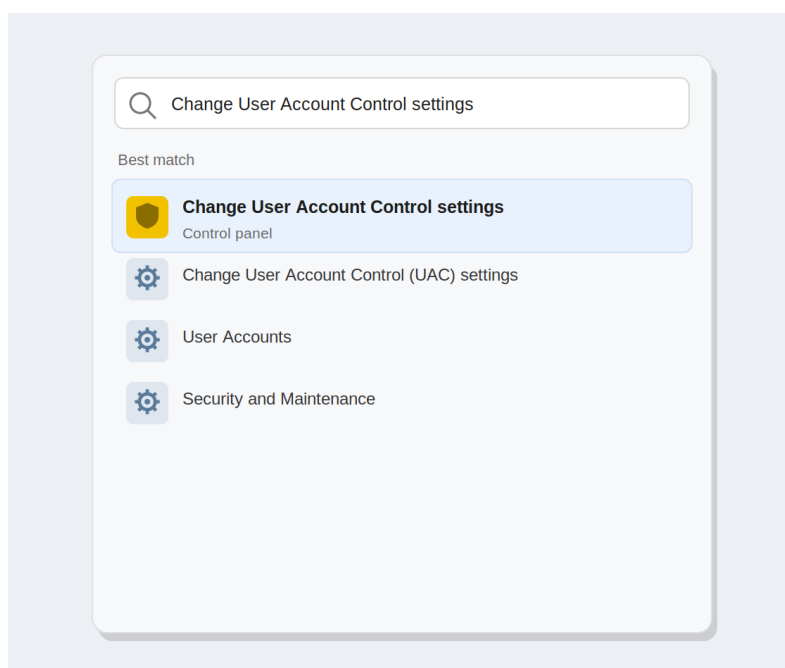
1. User account control
2. Applocker, smartscreen xavfsizlik tizimlari

**Diqqat:** Bu mashg'ulot amaliy bo'lib, 2-mashg'ulotda o'rgangan nazariy bilimni real (yoki virtual) Windows muhitida qo'lda sozlashga bag'ishlangan. Har bir amal maxsus o'quv laboratoriyasida yoki VirtualBox/VMware'dagi Windows virtual mashinasida bajarilishi kerak — hech qachon ishchi yoki shaxsiy kompyuterda sinab ko'rmang.

#### 1-savol. User Account Control (UAC)ni amalda sozlash

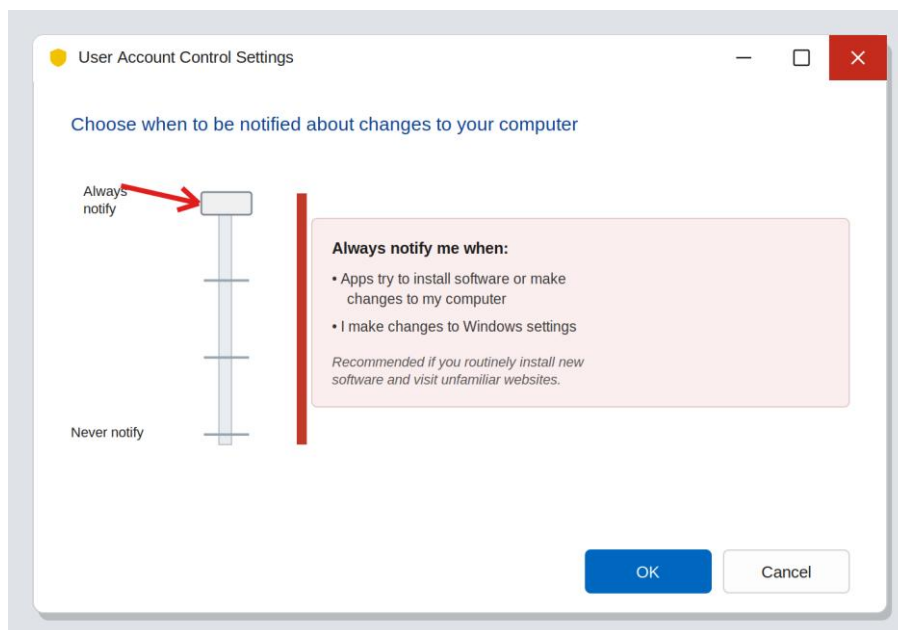
##### *Amaliy topshiriq 1.1: UAC darajasini o'zgartirish*

1. Boshlash menyusidan qidiruv qatoriga “User Account Control” deb yozing va “Change User Account Control settings” bo'limini oching.



17-rasm. Boshlash menyusida UAC sozlamasini qidirish

2. Ochilgan oynada chapdan o'ngga siljiydigan slider (sirg'anuvchi tugma) ko'rinadi — u to'rtta darajani ko'rsatadi.
3. Slider'ni “Always notify” (eng yuqori) holatiga o'tkazing va “OK” tugmasini bosing.



18-rasm. UAC slayderi eng yuqori “Always notify” holatida

4. Tizim UAC o‘zi ham tasdiqlashni so‘raydi — “Ha” deb javob bering.
5. Endi istalgan tizim sozlamasini o‘zgartirib ko‘ring (masalan, Sanani o‘zgartirish) — endi hatto Windows’ning o‘zi ham tasdiq so‘rayotganini kuzating.
6. Slider’ni yana standart (“Notify only when apps try to make changes”) holatiga qaytaring.

### ***Amaliy topshiriq 1.2: UACni buyruq qatori orqali tekshirish***

PowerShell’ni administrator huquqida oching va quyidagi buyruq bilan joriy UAC holatini tekshiring:

| Buyruq                                                                                                             | Natija                                                                       |
|--------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Get-ItemProperty<br>HKLM:\Software\Microsoft\Windows\CurrentVersion\Policy\System -Name EnableLUA                  | UAC yoqilgan (1) yoki o‘chirilgan (0) ekanini ko‘rsatadi                     |
| Get-ItemProperty<br>HKLM:\Software\Microsoft\Windows\CurrentVersion\Policy\System -Name ConsentPromptBehaviorAdmin | Administrator uchun joriy UAC darajasini raqam ko‘rinishida ko‘rsatadi (0–2) |

```
Windows PowerShell (Administrator)

Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\WINDOWS\system32> Get-ItemProperty HKLM:\Software\Microsoft\Windows\
CurrentVersion\Policies\System -Name EnableLUA

EnableLUA : 1
PSPath : Microsoft.PowerShell.Core\Registry::HKEY_LOCAL_MACHINE\...
PSProvider : Microsoft.PowerShell.Core\Registry

PS C:\WINDOWS\system32> _
```

19-rasm. PowerShell — UAC holatini (EnableLUA) tekshirish natijasi

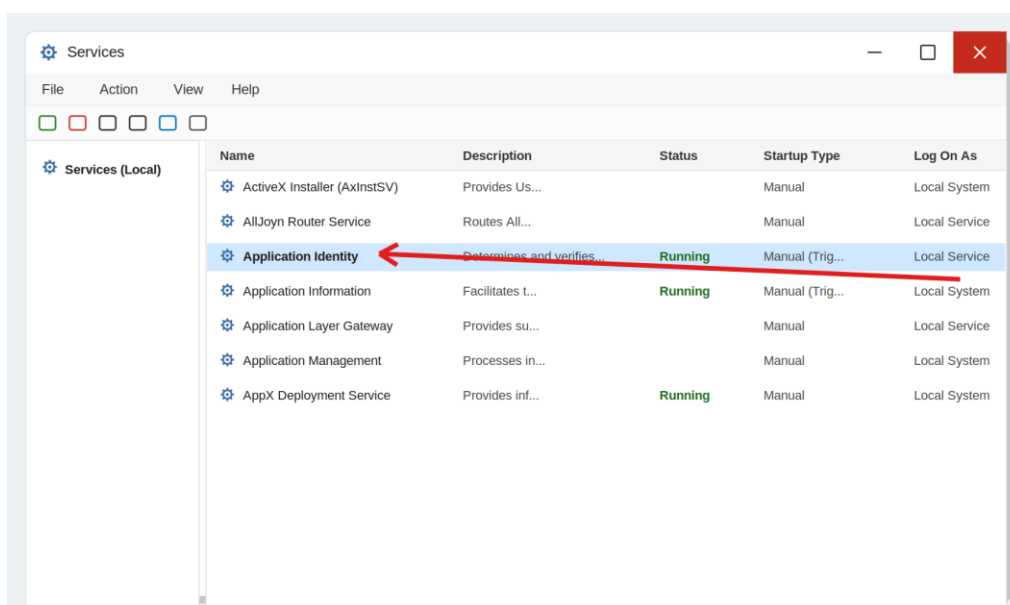
Natijada EnableLUA : 1 qiymati UAC yoqilganini bildiradi. Agar bu qiymat 0 bo'lsa, UAC o'chirilgan — bu jiddiy xavfsizlik kamchiligi hisoblanadi va darhol yoqilishi kerak. ConsentPromptBehaviorAdmin qiymati esa administrator uchun so'rov turini belgilaydi: 2 — “Ha/Yo‘q” so‘raladi (standart), 0 — hech narsa so‘ralmaydi (xavfli), 5 — dasturlar uchun so‘raladi.

*Eslatma:* bu registry kalitlarini qo‘lda o‘zgartirish mumkin, lekin GUI orqali o‘zgartirish tavsiya etiladi, chunki noto‘g‘ri qiymat tizim beqarorligiga olib kelishi mumkin.

## 2-savol. AppLocker va SmartScreen’ni amalda sozlash

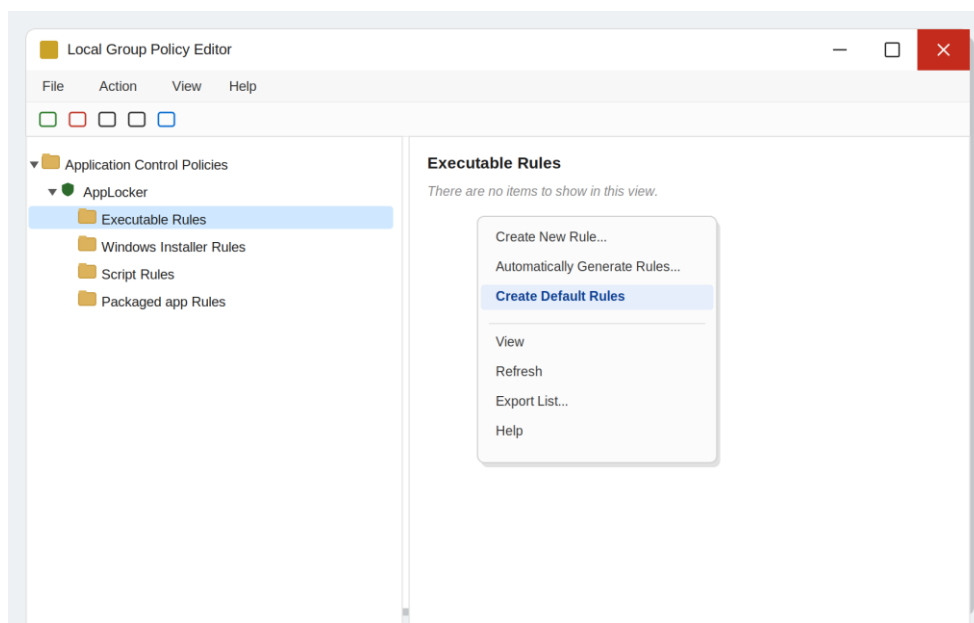
### Amaliy topshiriq 2.1: AppLocker’ni yoqish va qoida yaratish

1. Avvalo, kerakli servis ishlayotganini tekshiring: services.msc’ni oching, “Application Identity” servisini toping va uni “Automatic” qilib, ishga tushiring (1-mashg‘ulotda o‘rgangan bilim shu yerda amalda qo‘llanadi).



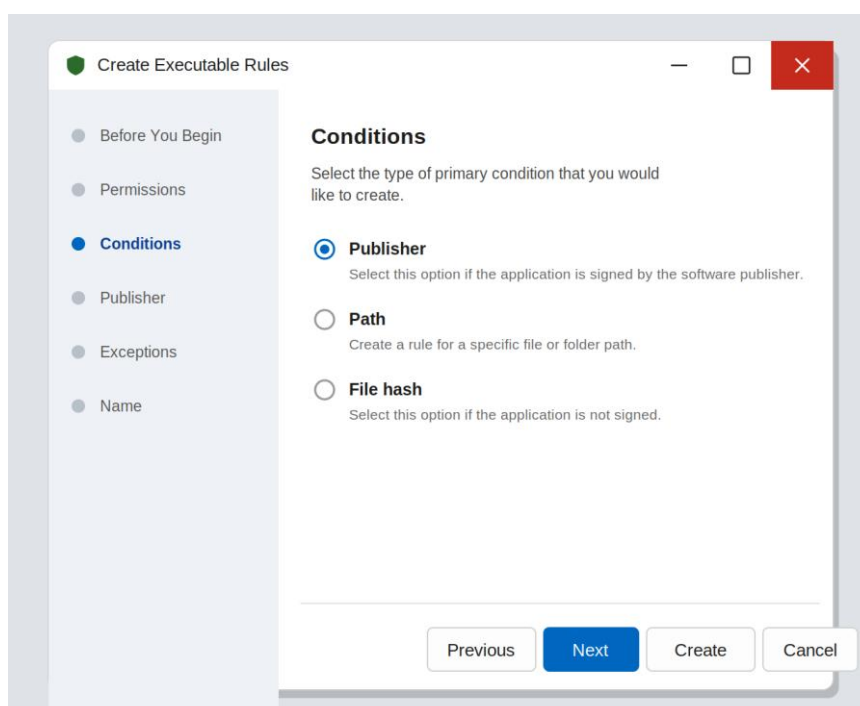
20-rasm. services.msc — “Application Identity” (AppIDSvc) servisi

2. **Win + R** → gpedit.msc buyrug‘i bilan Local Group Policy Editor’ni oching (faqat Pro/Enterprise nashrida mavjud).
3. Quyidagi yo‘lni bosib o‘ting: Computer Configuration → Windows Settings → Security Settings → Application Control Policies → AppLocker.
4. “Executable Rules” ustiga o‘ng tugma bosib, “Create Default Rules”ni tanlang — bu tizim uchun zarur bo‘lgan standart qoidalarni avtomatik yaratadi.



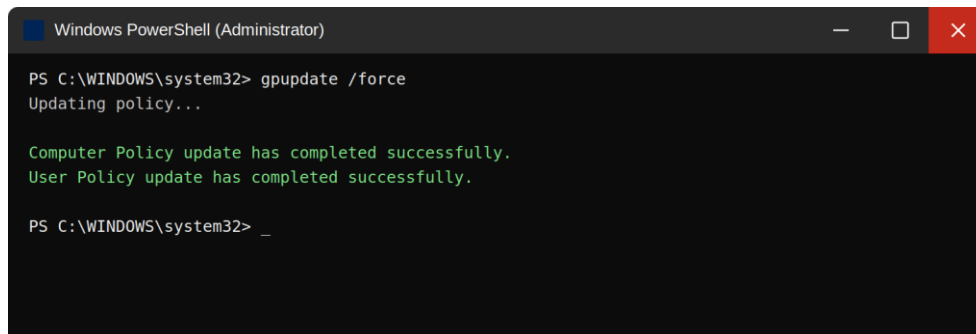
21-rasm. gpedit.msc — “Create Default Rules” buyrug‘i (kontekst menyusu)

5. Yangi qoida qo‘shish uchun yana o‘ng tugma → “Create New Rule” — so‘rovlarga javob berib, “Publisher” shartini tanlang va imzolangan dastur faylini (masalan, notepad.exe) ko‘rsating.



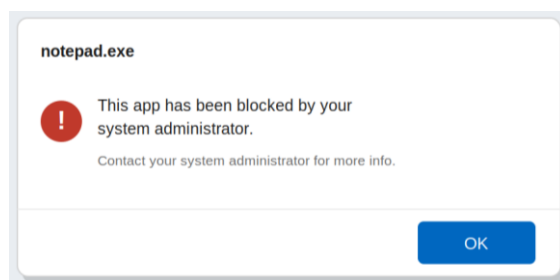
22-rasm. AppLocker — yangi qoida yaratish ustasi (Publisher sharti)

6. Qoidani “Allow” (ruxsat berish) rejimida saqlang.
7. AppLocker sozlamalarini kuchga kiritish uchun PowerShell’da: gpupdate /force buyrug‘ini bajaring.



23-rasm. PowerShell — “gpupdate /force” bilan siyosatni kuchga kiritish

8. Endi qoidada ko‘rsatilmagan biror dasturni ishga tushirishga urinib ko‘ring — u bloklanishi kerak.



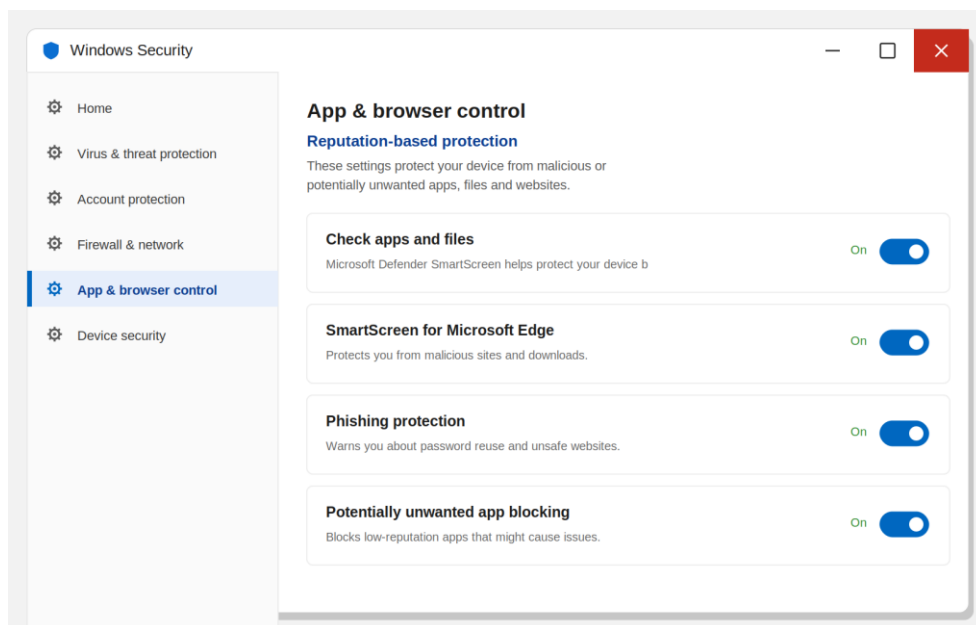
24-rasm. AppLocker tomonidan bloklangan dastur oynasi

### Tez-tez uchraydigan xatolar

- AppLocker qoidalari ishlamayapti — sababi ko‘pincha “Application Identity” servisi ishga tushirilmagan bo‘ladi.
- “Audit only” rejimi yoqilgan holda qoidalar hech narsani bloklamaydi — bu faqat kuzatish uchun mo‘ljallangan rejim, haqiqiy bloklash uchun “Enforce Rules” tanlanishi kerak.
- Standart qoidalar (Create Default Rules) yaratilmasa, tizimning o‘zi ishlamay qolishi mumkin — har doim avval standart qoidalarni yarating.

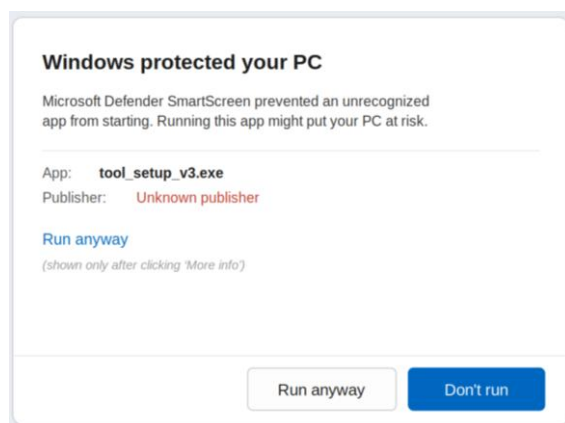
### Amaliy topshiriq 2.2: SmartScreen sozlamalarini tekshirish va sozlash

1. “Windows Security” dasturini oching → “App & browser control” bo‘limiga o‘ting.
2. “Reputation-based protection settings” havolasini bosing.



25-rasm. “App & browser control” — Reputation-based protection sozlamalari

3. Uchta asosiy sozlamani ko‘rasiz: “Check apps and files”, “SmartScreen for Microsoft Edge” va “Phishing protection” — barchasini “On” holatida qoldiring.
4. “Check apps and files” yoqilgan holda, internetdan (masalan, ishonchsiz test-sayt orqali) noma’lum kelib chiqishga ega bo‘lgan biror faylni yuklab ko‘ring — SmartScreen ogohlantirish oynasini ko‘rsatishini kuzating.
5. Ogohlantirish oynasidagi “More info” havolasini bosib, u qanday qo‘shimcha ma’lumot (nashriyotchi, obro‘ darajasi) ko‘rsatishini o‘rganing.



26-rasm. SmartScreen ogohlantirishi va “More info → Run anyway”

### Amaliy laboratoriya vazifasi (yakuniy)

Quyidagi vazifani virtual mashinada mustaqil bajaring va natijani hisobotda (screenshot bilan) taqdim eting:

- UAC darajasini “Always notify” holatiga o‘tkazing va buni tasdiqlovchi skrinshot oling.

- AppLocker orqali faqat C:\Windows va C:\Program Files papkalaridan dastur ishga tushishiga ruxsat beruvchi Path qoidasi yarating.
- Ushbu qoidadan tashqarida joylashgan istalgan .exe faylni ishga tushirishga urinib, uning bloklanganini skrinshot bilan tasdiqlang.
- SmartScreen yordamida noma'lum manbadan kelgan faylni yuklab, ogohlantirish oynasini skrinshot qiling.

### **Nazorat savollari — 3-mashg'ulot**

1. AppLocker ishlashi uchun qaysi servis ishga tushirilgan bo'lishi kerak?
2. "Audit only" va "Enforce Rules" rejimlari orasidagi farq nima?
3. UAC darajasini o'zgartirish uchun qaysi buyruq qatori vositasidan foydalanish mumkin?
4. SmartScreen ogohlantirishi chiqqanda foydalanuvchi qanday qo'shimcha ma'lumot ko'rishi mumkin?
5. Nima uchun standart AppLocker qoidalarini (Default Rules) yaratmasdan turib maxsus qoida qo'shish tavsiya etilmaydi?

## 4-mashg'ulot. Windows OT servislari (Amaliy)

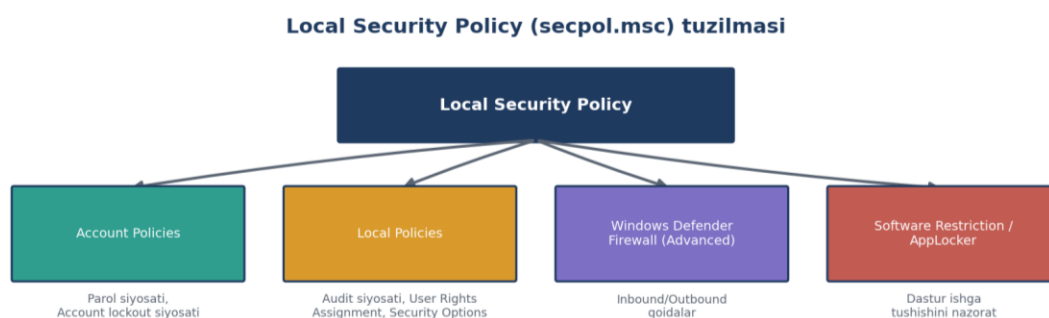
**Mavzu:** Windows OT xavfsizligi    **Mashg'ulot turi:** Amaliy    **Soat:** 4 soat

### O'quv savollari

1. Xavfsizlik sozlamalarini sozlash
2. Foydalanuvchilar xavfsizligini sozlash

### 1-savol. Xavfsizlik sozlamalarini sozlash (Local Security Policy)

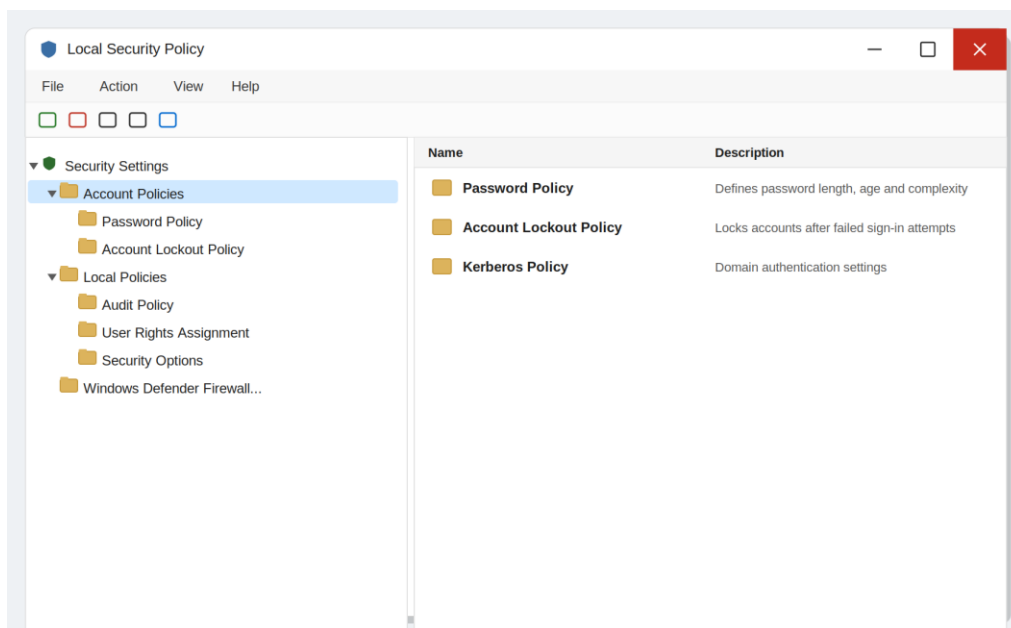
Windows'ning barcha muhim xavfsizlik sozlamalari (parol talablari, audit qoidalari, foydalanuvchi huquqlari) bitta markaziy joyda — Local Security Policy (secpol.msc) vositasida jamlangan. Bu vosita Windows Pro/Enterprise nashrlarida mavjud va uni domendagi kompyuterlarda Group Policy (gpedit.msc / Active Directory GPO) orqali ham markazlashtirilgan holda boshqarish mumkin.



#### Amaliy misollar (Security Options):

- Interactive logon: Do not display last user name
- Accounts: Rename administrator account
- Minimum password length / Password must meet complexity requirements
- Account lockout threshold — noto'g'ri urinishlar sonidan keyin hisobni bloklash

*27-rasm. Local Security Policy (secpol.msc) tuzilmasi*



28-rasm. Local Security Policy (secpol.msc) oynasi

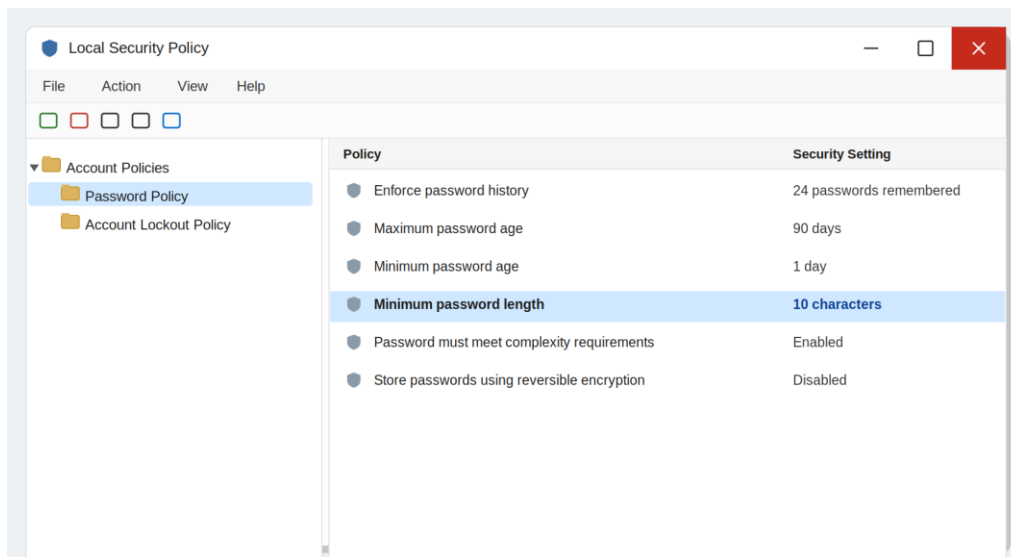
### 1.1. Account Policies (Hisob siyosati)

Bu bo‘lim parollar va hisoblarni bloklash qoidalarini belgilaydi. Xavfsiz tizim uchun tavsiya etiladigan qiymatlar:

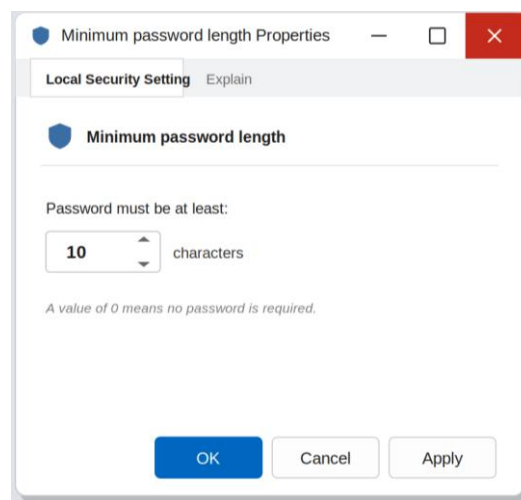
| Sozlama                                    | Tavsiya etiladigan qiymat   | Vazifasi                                                          |
|--------------------------------------------|-----------------------------|-------------------------------------------------------------------|
| Minimum password length                    | Kamida 8–12 belgi           | Qisqa, oson topiladigan parollarning oldini oladi                 |
| Password must meet complexity requirements | Yoqilgan (Enabled)          | Parolda katta/kichik harf, raqam va belgi bo‘lishini talab qiladi |
| Maximum password age                       | 60–90 kun                   | Parolni muntazam yangilashga majburlaydi                          |
| Account lockout threshold                  | 3–5 marta noto‘g‘ri urinish | Parolni “sinab ko‘rish” (brute-force) hujumidan himoya qiladi     |
| Account lockout duration                   | 15–30 daqiqa                | Bloklangan hisobning qancha vaqt yopiq turishini belgilaydi       |

#### Nega parol siyosati muhim?

- Agar parol atigi 6 ta belgidan iborat bo‘lsa, uni avtomatik “sinab ko‘rish” (brute-force) hujumi bir necha soatda topishi mumkin.
- 12 ta belgidan iborat, katta/kichik harf, raqam va belgidan tashkil topgan parolni topish esa amalda imkonsiz.
- Account lockout esa bunday urinishlarni 3–5 tadan keyin to‘xtatib, hujumni butunlay foydasiz qiladi.



29-rasm. Account Policies → Password Policy sozlamalari (secpol.msc)

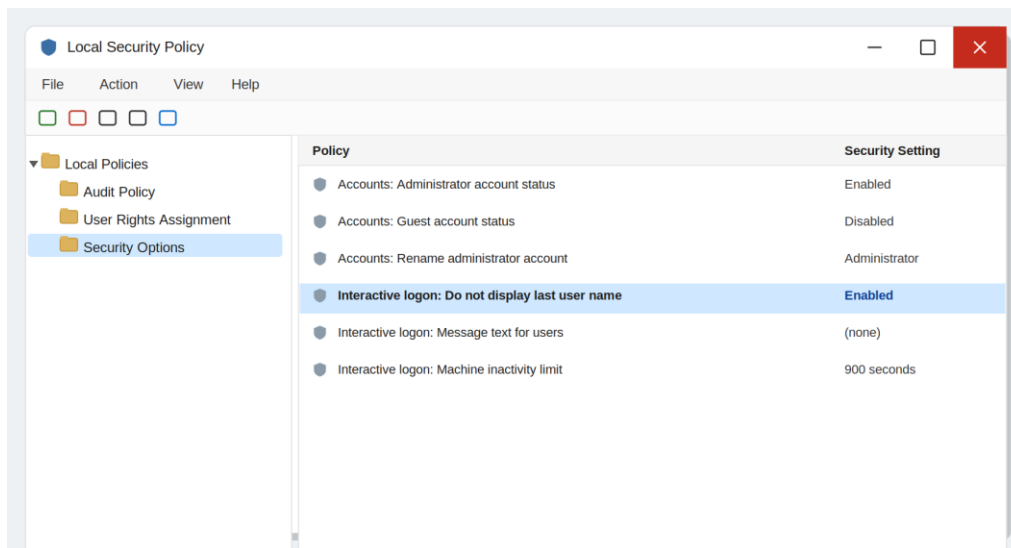


30-rasm. “Minimum password length” xususiyatlari oynasi

## 1.2. Local Policies — Audit va Security Options

Audit Policy — tizimda qanday hodisalar (login urinishlari, fayllarga kirish, sozlamalar o‘zgarishi) Windows Event Log‘ga yozib borilishini belgilaydi. Bu keyingi mavzularda o‘rganiladigan kiberkriminalistika va SIEM tizimlari uchun ma’lumot manbai bo‘lib xizmat qiladi. Security Options bo‘limida esa quyidagilar kabi muhim sozlamalar joylashgan:

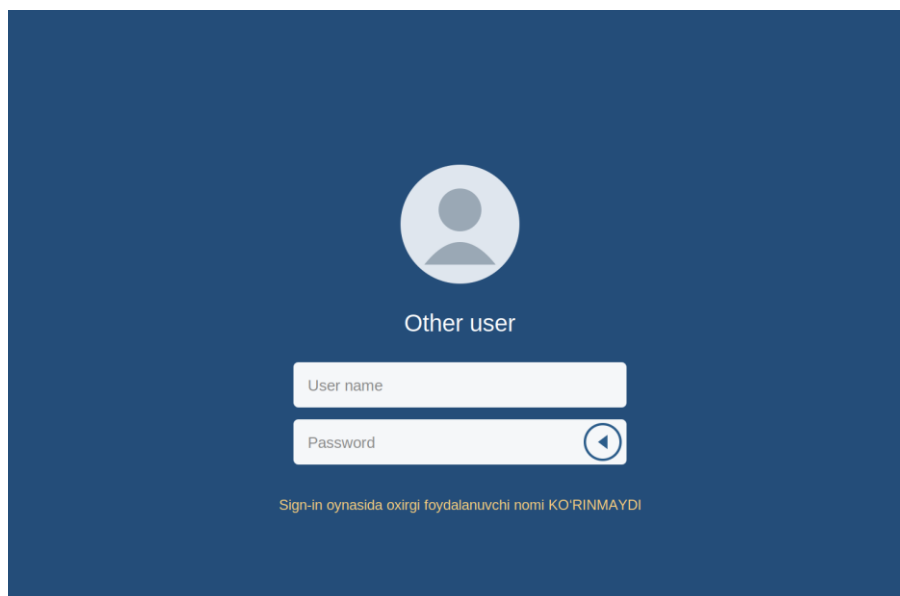
- **Interactive logon: Do not display last user name** — login oynasida oxirgi kirgan foydalanuvchi nomini yashiradi, bu buzg‘unchiga hisob nomini taxmin qilishni qiyinlashtiradi.
- **Accounts: Rename administrator account** — standart “Administrator” nomini o‘zgartirish, chunki bu nom hujumchilarga oldindan ma’lum bo‘ladi.
- **Accounts: Guest account status** — mehmon hisobini o‘chirib qo‘yish, chunki u ko‘pincha parolsiz yoki zaif himoyalangan bo‘ladi.



31-rasm. Local Policies → Security Options ro'yxati

### 1.3. Amaliy topshiriq: xavfsizlik sozlamasini o'zgartirish

1. **Win** + **R** → secpol.msc buyrug'ini bajaring.
2. Account Policies → Password Policy bo'limiga o'ting.
3. "Minimum password length" qiymatini 10 ga o'zgartiring.
4. Local Policies → Security Options bo'limiga o'ting.
5. "Interactive logon: Do not display last user name" sozlamasini toping va uni "Enabled" holatiga o'tkazing.
6. O'zgarishlarni saqlab, kompyuterni qulflab (**Win** + **L**) natijani tekshiring — endi login oynasida foydalanuvchi nomi ko'rinmasligi kerak.

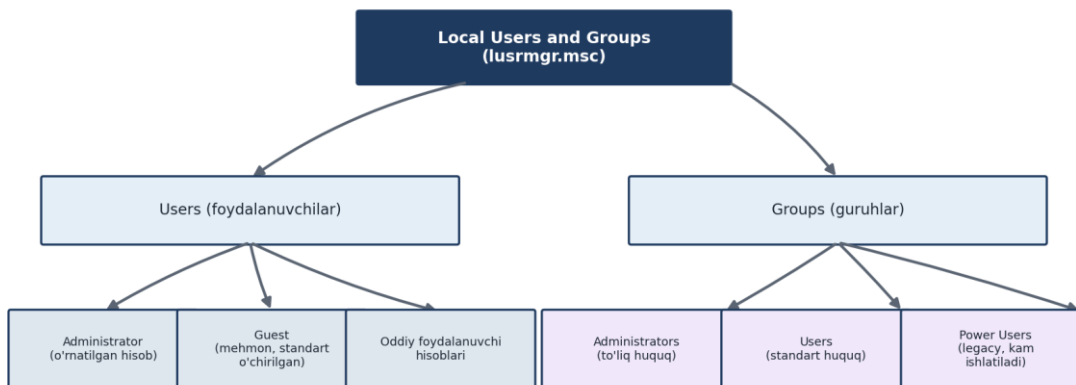


32-rasm. Natija: sign-in oynasida oxirgi foydalanuvchi nomi ko'rinmaydi

## 2-savol. Foydalanuvchilar xavfsizligini sozlash

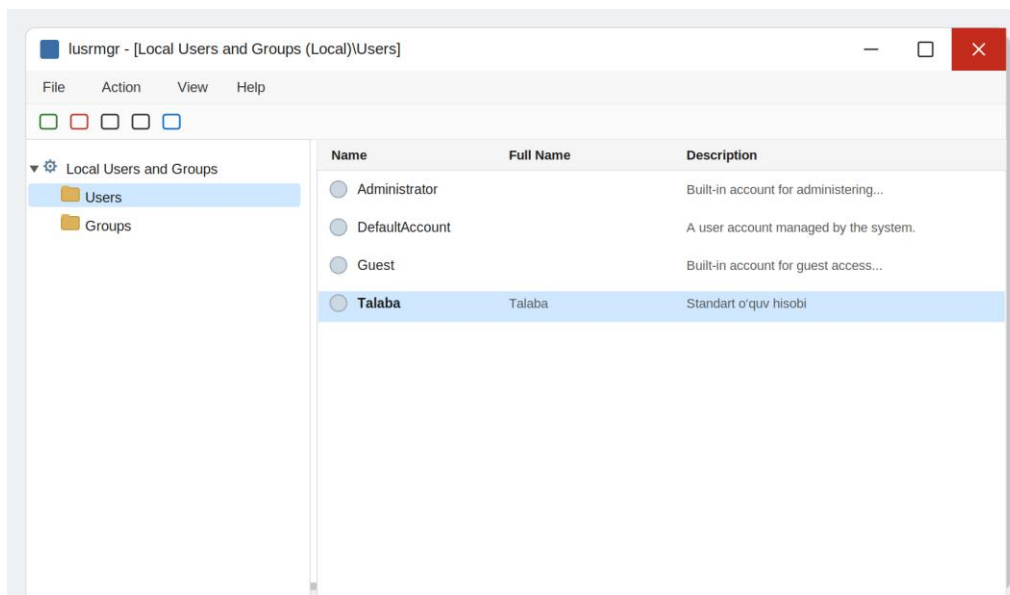
Windows'da har bir shaxs uchun alohida hisob (account) yaratiladi, va bu hisoblar guruhlariga (group) birlashtiriladi. Guruhga a'zolik orqali foydalanuvchiga qanday huquq berilishi belgilanadi — bu yondashuv “role-based access control” deb ataladi.

**Windows mahalliy foydalanuvchilar va guruhlar**



*Amaliy qoida: har bir foydalanuvchiga faqat ishi uchun zarur bo'lgan minimal huquq berish kerak (least privilege printsipli).*

*33-rasm. Windows mahalliy foydalanuvchilar va guruhlar*



*34-rasm. lusrmgr.msc — Local Users and Groups (foydalanuvchilar ro'yxati)*

| Guruh          | Huquq darajasi                                           | Tavsiya                                                       |
|----------------|----------------------------------------------------------|---------------------------------------------------------------|
| Administrators | To'liq — tizimning barcha sozlamalarini o'zgartira oladi | Faqat administrator vazifasini bajaruvchi shaxslarga berilsin |

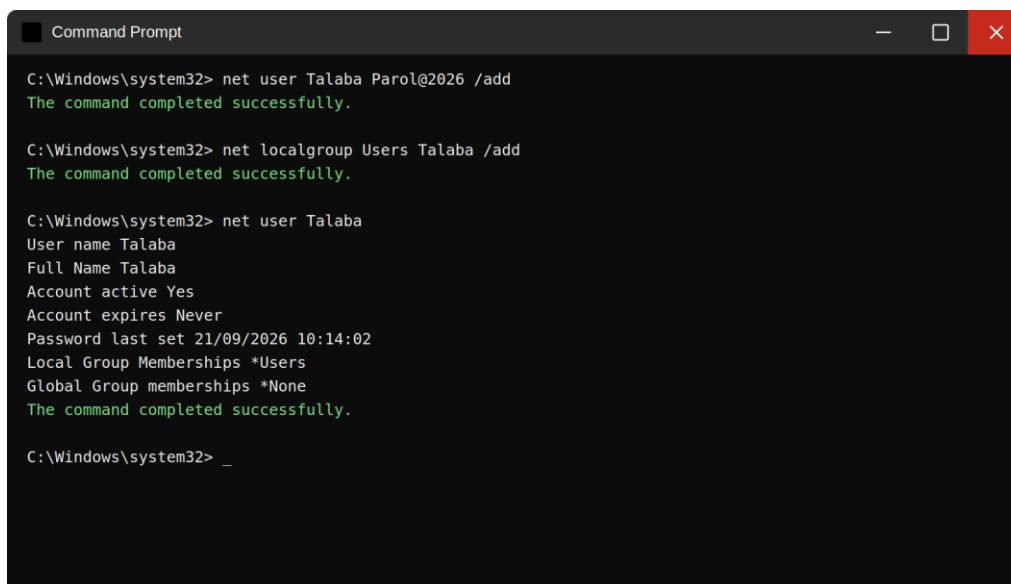
| Guruh  | Huquq darajasi                                                            | Tavsiya                                        |
|--------|---------------------------------------------------------------------------|------------------------------------------------|
| Users  | Standart — dastur oʻrnata olmaydi, tizim sozlamalarini oʻzgartira olmaydi | Kundalik ishchi hisoblar uchun tavsiya etiladi |
| Guests | Juda cheklangan, vaqtinchalik                                             | Iloji boricha oʻchirilgan holda saqlansin      |

### 2.1. Amaliy topshiriq: yangi cheklangan foydalanuvchi yaratish

GUI orqali (Computer Management → Local Users and Groups) yoki buyruq qatori orqali bajarish mumkin. Buyruq qatori usuli:

35-rasm. GUI usuli: “New User” oynasi orqali hisob yaratish

| Buyruq                                       | Vazifasi                                                                    |
|----------------------------------------------|-----------------------------------------------------------------------------|
| net user Talaba Parol@2026 /add              | “Talaba” nomli yangi hisob, berilgan parol bilan yaratadi                   |
| net localgroup Users Talaba /add             | Yangi hisobni standart “Users” guruhiga qoʻshadi (odatda avtomatik boʻladi) |
| net localgroup Administrators Talaba /delete | Agar hisob tasodifan Administrators guruhida boʻlsa, undan chiqaradi        |
| net user Talaba /active:yes                  | Hisobni faollashtiradi                                                      |
| net user Talaba                              | Hisob haqida toʻliq maʼlumotni (guruhlar, oxirgi login va h.k.) koʻrsatadi  |



```
Command Prompt

C:\Windows\system32> net user Talaba Parol@2026 /add
The command completed successfully.

C:\Windows\system32> net localgroup Users Talaba /add
The command completed successfully.

C:\Windows\system32> net user Talaba
User name Talaba
Full Name Talaba
Account active Yes
Account expires Never
Password last set 21/09/2026 10:14:02
Local Group Memberships *Users
Global Group memberships *None
The command completed successfully.

C:\Windows\system32> _
```

36-rasm. Buyruq qatori — “net user” bilan hisob yaratish va tekshirish

1. Yuqoridagi net user buyrug‘i bilan yangi, standart huquqli foydalanuvchi hisobini yarating.
2. Yangi hisob bilan tizimga qayta kiring (Sign out → yangi hisobni tanlang).
3. Ushbu hisob ostida biror dastur o‘rnatishga yoki tizim sozlamasini o‘zgartirishga urinib ko‘ring — UAC Credential prompt (login+parol so‘ralishi) chiqishini kuzating.
4. Administrator hisobiga qaytib, net user Talaba /delete buyrug‘i bilan test hisobini o‘chirib tashlang.

#### Xavfsizlik bo‘yicha yakuniy tavsiyalar (checklist)

- Har doim kundalik ishlar uchun standart (Users guruhidagi) hisobdan foydalaning, Administrator hisobini faqat zarurat bo‘lganda ishlating.
- Guest hisobini o‘chirib qo‘ying va standart Administrator nomini o‘zgartiring.
- UACni hech qachon “Never notify” holatiga o‘tkazmang.
- AppLocker yoki hech bo‘lmaganda SmartScreen’ni doimo yoqiq holda saqlang.
- Parol siyosatini (uzunlik, murakkablik, muddat) majburiy qiling va Account Lockout’ni faollashtiring.
- Windows Defender va Firewall’ni hech qachon o‘chirmang, hatto uchinchi tomon antivirusi o‘rnatilgan bo‘lsa ham ularning holatini tekshirib turing.

#### Nazorat savollari — 4-mashg‘ulot

1. Account lockout threshold va Account lockout duration sozlamalari nimani anglatadi?

2. Nega “Do not display last user name” sozlamasi xavfsizlikni oshiradi?
3. Administrators, Users va Guests guruhleri o‘rtasidagi farqni tushuntiring.
4. net user va net localgroup buyruqlarining vazifasini tushuntiring.
5. Nega administrator hisobi kundalik ishlar uchun ishlatilmasligi tavsiya etiladi?
6. “Role-based access control” tushunchasini o‘z so‘zlaringiz bilan tushuntiring.

## **Yakuniy xulosa**

Ushbu mavzuda Windows operatsion tizimining xavfsizligini ta'minlovchi asosiy mexanizmlar bosqichma-bosqich ko'rib chiqildi: avval fon jarayonlari — servislar — va ularning imtiyoz tizimi (1-mashg'ulot), so'ngra foydalanuvchi harakatlarini va dasturlarni nazorat qiluvchi vositalar — UAC, AppLocker, SmartScreen, Defender, Firewall (2-mashg'ulot), keyin esa shu bilimlarning amaliy qo'llanilishi — sozlash va konfiguratsiya qilish (3- va 4-mashg'ulotlar).

Bu bilimlar nafaqat mustaqil ahamiyatga ega, balki kursning keyingi mavzulari — virusologiya, tarmoq monitoring tizimlari (SIEM, IDS/IPS), va etikal haking (Metasploit, privilege escalation) — uchun ham zaruriy poydevor hisoblanadi. Xususan, roadmap.html faylida ko'rsatilgan yo'l xaritasi bo'yicha bu mavzu "Bosqich 02: Operatsion tizimlar — chuqur" bosqichiga to'g'ridan-to'g'ri mos keladi va "Bosqich 05: Xavfsizlik asoslari" hamda "Bosqich 08: Himoya (Blue Team)" bosqichlari uchun ham asos yaratadi.

## **Adabiyotlar va manbalar**

### ***Rasmiy hujjatlar (Microsoft)***

- Service Control Manager — <https://learn.microsoft.com/en-us/windows/win32/services/service-control-manager>
- About Services (Win32 apps) — <https://learn.microsoft.com/en-us/windows/win32/services/about-services>
- Windows System Services Fundamentals — <https://learn.microsoft.com/en-us/archive/technet-wiki/12229.windows-system-services-fundamentals>
- Get-Service / Start-Service / Stop-Service / Set-Service (PowerShell) — <https://learn.microsoft.com/en-us/powershell/module/microsoft.powershell.management/get-service>
- How User Account Control works — <https://learn.microsoft.com/en-us/windows/security/application-security/application-control/user-account-control/how-it-works>
- AppLocker overview — <https://learn.microsoft.com/en-us/windows/security/application-security/application-control/app-control-for-business/applocker/applocker-overview>

- Microsoft Defender SmartScreen overview — <https://learn.microsoft.com/en-us/windows/security/operating-system-security/virus-and-threat-protection/microsoft-defender-smartscreen/>
- Microsoft Defender Antivirus in Windows overview — <https://learn.microsoft.com/en-us/defender-endpoint/microsoft-defender-antivirus-windows>
- Windows Defender Firewall with Advanced Security (Explore) — <https://learn.microsoft.com/en-us/training/modules/manage-defender-windows-client/6-explore-windows-defender-firewall-advanced-security>

#### ***Xavfsizlik amaliyoti va hujum-texnikalari bo'yicha manbalar***

- MITRE ATT&CK — T1574.009: Hijack Execution Flow: Path Interception by Unquoted Path — <https://attack.mitre.org/techniques/T1574/009/>
- HackTricks — Windows Local Privilege Escalation Checklist — <https://hacktricks.wiki/en/windows-hardening/checklist-windows-privilege-escalation.html>

#### ***Kafedra tomonidan tavsiya etilgan asosiy adabiyot***

- Nishanov I.I., Mamajonov J.M. — “Kiberkriminalistika va etikal haking” fanining o'quv uslubiy majmuasi. Toshkent, 2025. — 390 b.
- Brian W. Kernighan. Understanding the Digital World: What You Need to Know about Computers, the Internet, Privacy, and Security, 2nd Edition. 2022.